# Additional Consensus Driven Pseudo-Labeling for Long-Tailed Multi-view Semi-Supervised Classification

Yuchen Zhang
*National University of Defense Technology*
Changsha, China
yuchenz0523@163.com

Hongbing Wu
*National University of Defense Technology*
Changsha, China
wuhongbing@nudt.edu.cn

Tingjin Luo*
*National University of Defense Technology*
Changsha, China
tingjinluo@hotmail.com

This document complements the main paper with the experimental results that are not included in the main body. Section I reports the complete Accuracy, G-mean, and Specificity tables for the same 84 experimental settings as the Macro-F1 table in the main paper, and Section II reports the full label-ratio results behind the label-ratio figure in the main paper. Sections III–VIII provide the additional figures for the radar charts, pseudo-label analysis, class-group comparison, parameter sensitivity, and representation visualization, together with the extended results on the clinical IGF-1 application. All conventions, datasets, comparison methods, and evaluation metrics follow the definitions in Section IV of the main paper. Throughout, *C*/*I*/*A*/*U* denote the consistent, inverse, arbitrary, and uniform unlabeled distributions, respectively, and $IR$ and $LR$ denote the imbalance ratio and the labeling ratio.

## I. Additional Full Results Tables

The main paper reports the complete Macro-F1 results across all seven benchmarks under $IR \in \{0.1, 0.2, 0.3\}$ and the four unlabeled distributions at $LR = 0.09$. This section provides the corresponding full tables for the three remaining evaluation metrics: Accuracy (Table I), G-mean (Table II), and macro-averaged Specificity (Table III). In each table, the best mean is pink and shown in bold, while the second-best mean is blue and underlined. Values are reported as mean $\pm$ standard deviation in percent. DSRL and GEGCN ran out of memory on CIFAR-10, CIFAR-100, NoisyMNIST, and YouTubeFace20; DSRL additionally ran out of memory on the three OutScene uniform settings. Missing runs are excluded from the ranking.

## II. Full Label-Ratio Results

The main paper illustrates the impact of the labeling ratio ($LR$) on Macro-F1 across CIFAR-100, GRAZ02, NoisyMNIST, and OutScene. Tables IV–VII provide the complete results for four evaluation metrics at $LR \in \{0.03, 0.06, 0.09, 0.12, 0.15\}$ under the consistent long-tailed setting ($IR = 0.1$). Results are presented as mean $\pm$ standard deviation across three random seeds, with missing results due to memory limitations denoted by "$-$". Method names include the selected learning rates, and CDPL is denoted as *Ours*. CDPL consistently achieves the highest Macro-F1 across all datasets and labeling ratios, with increasingly pronounced gains over the strongest baseline as the labeling ratio decreases.

*Corresponding author.

## III. Additional Radar Charts at $IR = 0.2$ and $IR = 0.3$

The main paper visualizes the Accuracy, G-mean, and Specificity of all compared methods at $IR = 0.1$. Figures 1 and 2 extend the comparison to $IR = 0.2$ and $IR = 0.3$, respectively, using the same radar representation. Consistent with the main paper, the CDPL polygon remains the outermost on all six axes and in all panels, confirming that the advantage of CDPL is not specific to the most severe imbalance setting.

## IV. Additional Pseudo-Label Analysis

The main paper reports the pseudo-label error rate and utilization rate on CIFAR-100, NoisyMNIST, and YouTube-Face20, together with the row-normalized confusion matrices of SimPro and CDPL from epoch 100 to 500. This section supplements those results with (i) the pseudo-label statistics on CIFAR-10 (Figure 3) and (ii) the row-normalized pseudo-label confusion matrices of SimPro and CDPL on NoisyMNIST at $LR = 0.09$ and $IR = 0.1$ (Figure 4).

## V. Additional Class-Group Results

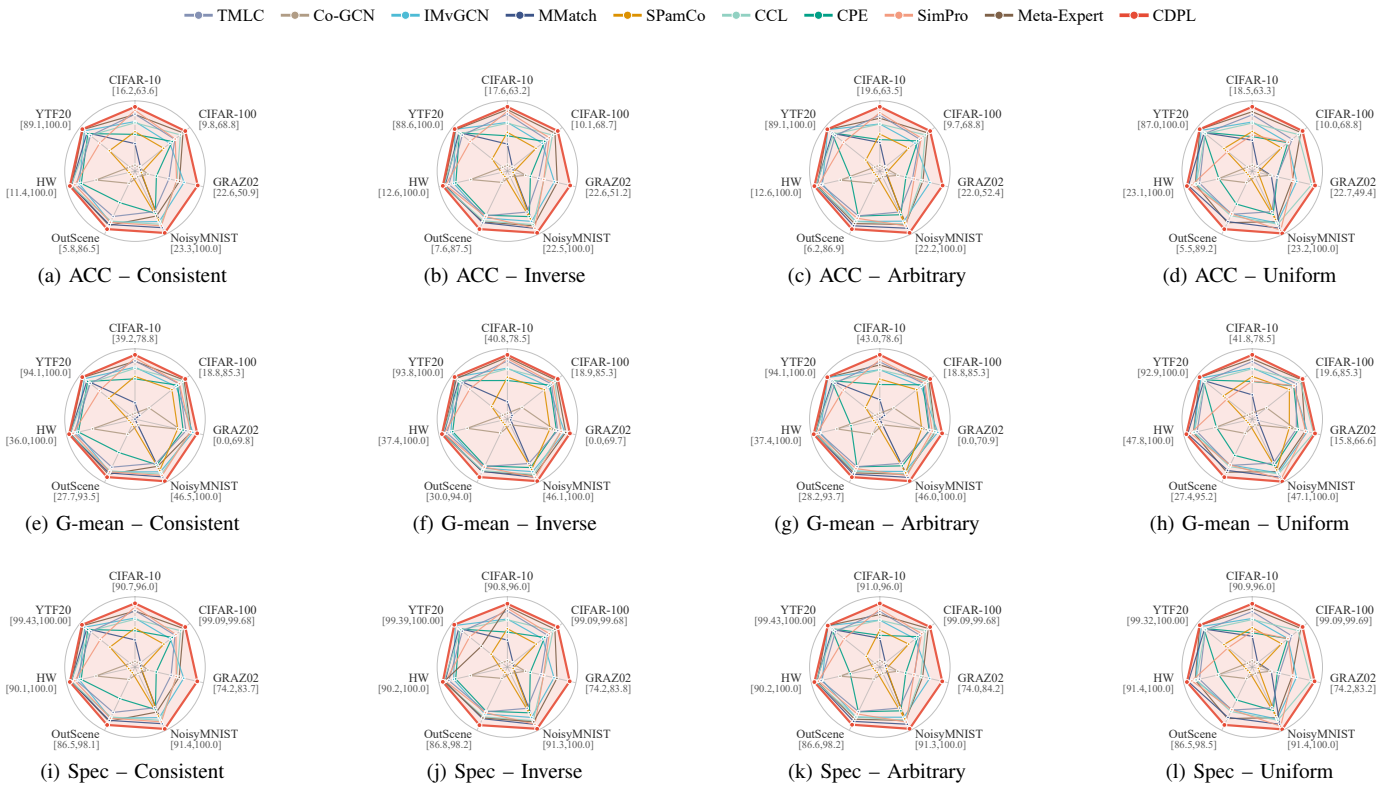
The main paper evaluates Macro-F1 across head, medium, and tail classes on CIFAR-10, NoisyMNIST, CIFAR-100, and YouTubeFace20. Figure 5 extends this analysis to Handwritten and OutScene, where CDPL consistently achieves the highest Macro-F1 across all class groups, with the most pronounced gains on tail classes.

## VI. Additional Parameter Sensitivity

The main paper analyzes the sensitivity of CDPL to the trade-off coefficients $\lambda_1$, $\lambda_2$, and $\lambda_3$ on NoisyMNIST and OutScene. Figure 6 extends this analysis to Handwritten and GRAZ02. The results are consistent with the main paper: performance is relatively insensitive to $\lambda_1$ and is best when $\lambda_2$ is near 10 and $\lambda_3$ is near 1.

## VII. Comprehensive t-SNE Visualization of CDPL Representations

The main paper visualizes nine methods on the YouTube-Face20 test set. Figures 7–8 further examine CDPL's fused representations on the NoisyMNIST and YouTubeFace20 datasets

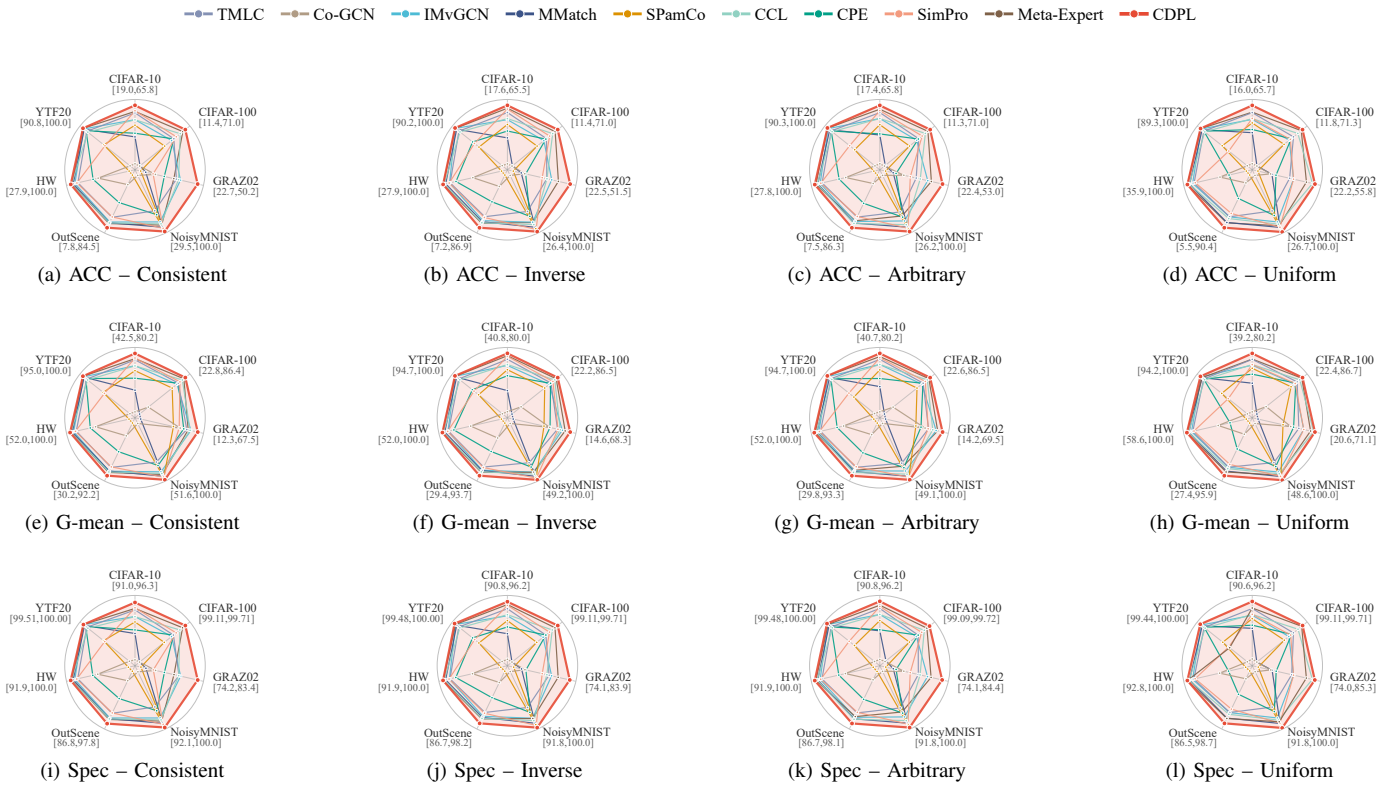



(a) ACC – Consistent (b) ACC – Inverse (c) ACC – Arbitrary (d) ACC – Uniform

(e) G-mean – Consistent (f) G-mean – Inverse (g) G-mean – Arbitrary (h) G-mean – Uniform

(i) Spec – Consistent (j) Spec – Inverse (k) Spec – Arbitrary (l) Spec – Uniform

Fig. 1. Accuracy, G-mean, and Specificity of all compared methods on six datasets under the four unlabeled distributions at $IR = 0.2$ and $LR = 0.09$.




(a) ACC – Consistent (b) ACC – Inverse (c) ACC – Arbitrary (d) ACC – Uniform

(e) G-mean – Consistent (f) G-mean – Inverse (g) G-mean – Arbitrary (h) G-mean – Uniform

(i) Spec – Consistent (j) Spec – Inverse (k) Spec – Arbitrary (l) Spec – Uniform

Fig. 2. Accuracy, G-mean, and Specificity of all compared methods on six datasets under the four unlabeled distributions at $IR = 0.3$ and $LR = 0.09$.

TABLE I

ACCURACY (%) ON SEVEN DATASETS AT $LR = 0.09$ AND $IR \in \{0.1, 0.2, 0.3\}$ UNDER FOUR UNLABELED DISTRIBUTIONS. VALUES ARE MEAN ± REPORTED VARIATION. THE BEST AND SECOND-BEST MEANS ARE MARKED BY PINK/BOLD AND BLUE/UNDERLINED ENTRIES, RESPECTIVELY. THE Δ ROW REPORTS CDPL MINUS THE STRONGEST COMPETING METHOD (PERCENTAGE POINTS); POSITIVE/NEGATIVE DIFFERENCES ARE BOLD RED/BLUE.

| Dataset | Method | $IR = 0.1$ | | | | $IR = 0.2$ | | | | $IR = 0.3$ | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | C | I | A | U | C | I | A | U | C | I | A | U |
| **CIFAR-100** | CCL | 52.11±0.006 | 51.65±0.005 | 51.23±0.002 | 52.55±0.006 | 58.61±0.004 | 58.29±0.002 | 58.58±0.007 | 59.25±0.010 | 60.49±0.004 | 60.54±0.004 | 60.79±0.008 | 61.16±0.015 |
| | CPE | 42.91±0.009 | 41.02±0.009 | 43.70±0.003 | 44.01±0.009 | 49.75±0.003 | 49.91±0.011 | 50.50±0.010 | 48.20±0.010 | 53.37±0.008 | 53.10±0.002 | 53.18±0.021 | 54.02±0.004 |
| | SimPro | 44.87±0.014 | 47.47±0.005 | 47.46±0.018 | 40.84±0.036 | 53.04±0.016 | 54.76±0.009 | 54.22±0.012 | 48.78±0.011 | 56.27±0.014 | 57.19±0.010 | 57.28±0.011 | 53.07±0.017 |
| | Meta-Expert | 52.29±0.030 | 49.78±0.018 | 52.29±0.016 | 52.24±0.003 | 61.37±0.005 | 60.84±0.005 | 61.29±0.009 | 61.96±0.006 | 63.30±0.001 | 63.21±0.013 | 63.61±0.006 | 64.91±0.004 |
| | Co-GCN | 12.47±0.084 | 13.03±0.082 | 12.66±0.083 | 12.65±0.081 | 14.72±0.105 | 15.06±0.099 | 14.63±0.102 | 14.93±0.095 | 16.41±0.109 | 16.40±0.108 | 16.31±0.109 | 16.84±0.113 |
| | IMvGCN | 40.43±0.003 | 41.39±0.004 | 41.21±0.004 | 42.79±0.002 | 47.22±0.007 | 47.30±0.001 | 47.52±0.003 | 48.20±0.001 | 51.17±0.004 | 50.97±0.002 | 51.44±0.003 | 51.91±0.004 |
| | MMatch | 13.72±0.001 | 14.07±0.001 | 14.11±0.000 | 14.00±0.000 | 15.64±0.000 | 15.46±0.000 | 15.62±0.004 | 15.86±0.003 | 16.86±0.005 | 16.81±0.001 | 17.23±0.003 | 16.91±0.009 |
| | SPamCo | 23.89±0.127 | 32.35±0.003 | 31.97±0.003 | 32.74±0.010 | 40.59±0.004 | 40.59±0.004 | 40.59±0.004 | 41.23±0.003 | 43.61±0.004 | 43.61±0.004 | 43.61±0.004 | 46.28±0.002 |
| | TMLC | 47.11±0.000 | 47.34±0.001 | 47.41±0.000 | 46.89±0.001 | 52.62±0.000 | 52.35±0.000 | 52.71±0.001 | 52.68±0.005 | 53.57±0.001 | 53.77±0.000 | 53.56±0.000 | 56.48±0.001 |
| | CDPL | **58.94±0.003** | **58.94±0.003** | **58.94±0.003** | **59.61±0.002** | **63.82±0.004** | **63.82±0.004** | **63.82±0.004** | **63.89±0.004** | **66.00±0.006** | **66.00±0.006** | **66.00±0.006** | **66.26±0.002** |
| | Δ | **+6.65** | **+7.29** | **+6.65** | **+7.06** | **+2.45** | **+2.98** | **+2.53** | **+1.93** | **+2.70** | **+2.79** | **+2.39** | **+1.35** |
| **CIFAR-10** | CCL | 41.83±0.022 | 43.01±0.033 | 42.83±0.030 | 43.17±0.021 | 48.81±0.030 | 48.73±0.030 | 49.05±0.033 | 49.16±0.029 | 52.26±0.044 | 52.48±0.046 | 52.69±0.048 | 52.17±0.047 |
| | CPE | 36.69±0.014 | 35.26±0.018 | 36.10±0.009 | 36.96±0.005 | 41.00±0.011 | 40.52±0.004 | 39.30±0.016 | 40.51±0.012 | 43.32±0.009 | 44.01±0.011 | 42.08±0.006 | 44.48±0.016 |
| | SimPro | 53.71±0.042 | 54.07±0.028 | 53.69±0.031 | 53.74±0.032 | 57.28±0.039 | 56.46±0.033 | 56.15±0.029 | 41.50±0.143 | 58.29±0.026 | 59.00±0.032 | 58.12±0.024 | 52.34±0.095 |
| | Meta-Expert | 43.55±0.139 | 54.27±0.040 | 45.28±0.138 | 47.01±0.149 | 54.41±0.061 | 57.53±0.028 | 52.66±0.092 | 56.40±0.041 | 57.85±0.057 | 59.54±0.031 | 59.09±0.027 | 56.76±0.066 |
| | Co-GCN | 19.70±0.057 | 21.99±0.034 | 20.53±0.046 | 20.82±0.051 | 20.23±0.061 | 21.47±0.051 | 23.28±0.036 | 22.24±0.039 | 22.96±0.043 | 21.61±0.051 | 21.50±0.053 | 20.22±0.054 |
| | IMvGCN | 45.44±0.002 | 44.79±0.004 | 44.76±0.004 | 46.31±0.002 | 49.64±0.004 | 49.25±0.003 | 49.06±0.002 | 49.84±0.007 | 52.36±0.002 | 51.76±0.003 | 52.70±0.001 | 51.61±0.002 |
| | MMatch | 29.06±0.011 | 29.90±0.011 | 29.45±0.086 | 36.63±0.015 | 34.78±0.027 | 35.16±0.033 | 37.49±0.089 | 38.23±0.048 | 40.89±0.009 | 39.33±0.012 | 41.30±0.046 | 42.21±0.005 |
| | SPamCo | 36.21±0.004 | 35.63±0.002 | 35.62±0.002 | 36.70±0.000 | 42.78±0.005 | 42.78±0.005 | 42.78±0.005 | 43.81±0.001 | 48.67±0.005 | 48.46±0.005 | 48.46±0.005 | 49.11±0.004 |
| | TMLC | 53.18±0.000 | 52.98±0.000 | 52.96±0.001 | 52.08±0.000 | 54.64±0.001 | 54.68±0.000 | 54.61±0.002 | 54.41±0.001 | 56.88±0.000 | 56.61±0.000 | 56.80±0.002 | 56.40±0.000 |
| | CDPL | **54.58±0.002** | **55.88±0.000** | **54.97±0.008** | **55.44±0.001** | **59.57±0.006** | **59.37±0.002** | **59.81±0.002** | **59.54±0.005** | **61.90±0.004** | **61.51±0.003** | **61.73±0.001** | **61.51±0.004** |
| | Δ | **+0.87** | **+1.61** | **+1.28** | **+1.70** | **+2.29** | **+1.84** | **+3.66** | **+3.14** | **+3.61** | **+1.97** | **+2.64** | **+4.75** |
| **GRAZ02** | CCL | 35.75±0.018 | 34.27±0.015 | 32.80±0.006 | 44.62±0.047 | 39.92±0.008 | 37.90±0.011 | 40.32±0.015 | 45.83±0.038 | 40.19±0.005 | 39.25±0.008 | 40.46±0.006 | 51.75±0.008 |
| | CPE | 31.45±0.091 | 28.23±0.046 | 31.51±0.029 | 34.75±0.027 | 31.45±0.091 | 32.20±0.047 | 33.57±0.033 | 32.46±0.106 | 34.46±0.123 | 30.24±0.074 | 29.66±0.088 | 33.86±0.114 |
| | SimPro | 38.84±0.030 | 39.38±0.023 | 38.44±0.012 | 43.01±0.019 | 39.78±0.026 | 37.77±0.054 | 37.23±0.030 | 38.58±0.026 | 29.97±0.009 | 37.23±0.012 | 36.02±0.023 | 42.47±0.005 |
| | Meta-Expert | 37.23±0.034 | 36.42±0.013 | 36.42±0.031 | 42.88±0.002 | 40.99±0.033 | 43.41±0.023 | 41.13±0.058 | 38.31±0.057 | 38.31±0.023 | 44.09±0.008 | 45.43±0.043 | 48.79±0.023 |
| | Co-GCN | 29.03±0.033 | 28.16±0.048 | 25.47±0.037 | 27.55±0.033 | 27.96±0.038 | 30.24±0.040 | 30.04±0.035 | 29.64±0.038 | 30.91±0.037 | 29.84±0.014 | 33.13±0.025 | 31.18±0.045 |
| | DSRL | 29.03±0.000 | 27.02±0.000 | 25.00±0.000 | 35.48±0.000 | 31.05±0.000 | 31.85±0.000 | 25.00±0.000 | 33.87±0.000 | 29.84±0.000 | 25.00±0.000 | 29.44±0.000 | 37.50±0.000 |
| | GEGCN | 26.48±0.026 | 26.34±0.022 | 27.96±0.030 | 34.95±0.023 | 31.05±0.011 | 29.70±0.043 | 31.59±0.076 | 37.77±0.015 | 40.86±0.019 | 39.11±0.054 | 38.31±0.032 | 43.01±0.093 |
| | IMvGCN | 38.44±0.010 | 40.73±0.007 | 40.19±0.019 | 41.53±0.019 | 42.88±0.014 | 42.20±0.010 | 44.22±0.014 | 40.05±0.016 | 40.99±0.008 | 41.26±0.008 | 43.82±0.033 | 42.47±0.014 |
| | MMatch | 24.87±0.002 | 25.00±0.000 | 25.00±0.000 | 29.03±0.000 | 25.00±0.000 | 25.27±0.002 | 24.60±0.000 | 30.11±0.002 | 27.02±0.004 | 28.36±0.013 | 28.36±0.008 | 31.85±0.004 |
| | SPamCo | 25.00±0.000 | 25.00±0.000 | 25.00±0.000 | 25.00±0.000 | 25.27±0.002 | 25.00±0.000 | 25.00±0.000 | 25.00±0.000 | 25.00±0.000 | 25.00±0.000 | 25.00±0.000 | 25.00±0.000 |
| | TMLC | 32.88±0.013 | 35.30±0.033 | 30.99±0.013 | 31.94±0.047 | 36.64±0.003 | 35.70±0.006 | 36.51±0.029 | 32.34±0.003 | 39.33±0.005 | 39.19±0.004 | 38.66±0.024 | 40.54±0.022 |
| | CDPL | **42.88±0.024** | **49.19±0.029** | **46.64±0.018** | **47.85±0.021** | **48.52±0.031** | **48.79±0.025** | **49.87±0.024** | **47.18±0.011** | **47.85±0.020** | **49.06±0.012** | **50.40±0.008** | **52.96±0.015** |
| | Δ | **+4.04** | **+8.46** | **+6.45** | **+3.23** | **+5.64** | **+5.38** | **+5.65** | **+1.35** | **+6.86** | **+4.97** | **+4.97** | **+1.21** |
| **Handwritten** | CCL | 91.08±0.018 | 91.17±0.019 | 92.08±0.008 | 88.00±0.015 | 94.25±0.007 | 93.67±0.008 | 94.00±0.005 | 86.00±0.012 | 94.67±0.005 | 94.42±0.004 | 93.67±0.008 | 94.83±0.001 |
| | CPE | 50.25±0.149 | 59.75±0.279 | 61.00±0.279 | 46.50±0.516 | 80.88±0.140 | 78.38±0.076 | 49.88±0.507 | 60.50±0.286 | 72.00±0.152 | 82.00±0.149 | 70.75±0.301 | 57.62±0.394 |
| | SimPro | 81.75±0.013 | 83.92±0.018 | 82.08±0.032 | 78.92±0.089 | 89.08±0.047 | 91.42±0.038 | 90.92±0.039 | 90.83±0.021 | 89.00±0.074 | 89.17±0.040 | 90.00±0.035 | 89.17±0.081 |
| | Meta-Expert | 91.33±0.008 | 90.75±0.026 | 89.50±0.030 | 91.00±0.023 | 93.42±0.010 | 93.33±0.005 | 92.50±0.004 | 87.00±0.003 | 94.25±0.013 | 95.42±0.004 | 95.17±0.005 | 96.50±0.009 |
| | Co-GCN | 62.67±0.245 | 59.38±0.253 | 58.83±0.280 | 59.50±0.218 | 61.67±0.244 | 59.46±0.242 | 63.08±0.239 | 62.08±0.220 | 66.67±0.174 | 64.29±0.221 | 63.75±0.213 | 66.08±0.187 |
| | DSRL | 56.50±0.000 | 48.50±0.000 | 51.00±0.000 | 54.75±0.000 | 61.25±0.000 | 58.50±0.000 | 59.25±0.000 | 59.50±0.000 | 67.50±0.000 | 64.25±0.000 | 64.25±0.000 | 64.50±0.000 |
| | GEGCN | 78.00±0.208 | 85.92±0.086 | 67.08±0.339 | 87.92±0.073 | 64.25±0.282 | 82.75±0.138 | 92.25±0.049 | 67.50±0.280 | 79.33±0.146 | 92.67±0.056 | 89.83±0.116 | 70.92±0.110 |
| | IMvGCN | 80.33±0.016 | 79.00±0.005 | 80.83±0.023 | 82.17±0.006 | 83.33±0.013 | 81.00±0.013 | 85.83±0.024 | 86.50±0.005 | 91.17±0.003 | 88.83±0.015 | 90.67±0.003 | 91.67±0.013 |
| | MMatch | 86.12±0.012 | 84.75±0.028 | 86.75±0.032 | 88.62±0.009 | 93.75±0.011 | 92.00±0.014 | 93.75±0.000 | 94.75±0.004 | 92.88±0.002 | 92.88±0.002 | 93.12±0.009 | 94.50±0.004 |
| | SPamCo | 21.67±0.001 | 21.67±0.001 | 21.67±0.001 | 21.75±0.003 | 19.08±0.027 | 20.17±0.009 | 20.25±0.008 | 29.83±0.017 | 34.08±0.010 | 34.08±0.010 | 34.08±0.010 | 41.42±0.008 |
| | TMLC | 83.92±0.017 | 84.00±0.020 | 83.75±0.003 | 82.83±0.007 | 87.17±0.000 | 87.25±0.008 | 87.33±0.002 | 82.25±0.005 | 88.00±0.009 | 88.17±0.002 | 88.67±0.001 | 90.33±0.001 |
| | CDPL | **92.42±0.010** | **94.33±0.005** | **93.50±0.007** | **95.25±0.013** | **95.58±0.003** | **95.00±0.007** | **96.08±0.003** | **96.42±0.001** | **95.50±0.004** | **95.58±0.014** | **96.33±0.005** | **96.58±0.005** |
| | Δ | **+1.09** | **+3.16** | **+1.42** | **+4.25** | **+1.33** | **+1.33** | **+2.08** | **+1.67** | **+0.83** | **+0.16** | **+1.16** | **+0.08** |
| **NoisyMNIST** | CCL | 80.83±0.003 | 81.31±0.003 | 81.37±0.005 | 81.53±0.004 | 87.71±0.001 | 87.95±0.002 | 87.54±0.005 | 87.50±0.002 | 89.75±0.004 | 90.16±0.004 | 90.22±0.002 | 89.69±0.003 |
| | CPE | 66.97±0.017 | 68.57±0.012 | 65.79±0.013 | 68.73±0.006 | 74.46±0.002 | 78.18±0.004 | 75.76±0.016 | 75.74±0.014 | 79.50±0.004 | 80.54±0.014 | 81.43±0.033 | 79.67±0.006 |
| | SimPro | 86.17±0.051 | 89.46±0.025 | 89.64±0.007 | 90.00±0.012 | 89.59±0.003 | 92.44±0.030 | 89.62±0.008 | 94.90±0.002 | 94.66±0.021 | 95.76±0.005 | 94.36±0.015 | 93.28±0.051 |
| | Meta-Expert | 75.03±0.126 | 82.04±0.104 | 82.78±0.074 | 81.45±0.138 | 77.63±0.236 | 89.37±0.016 | 87.70±0.032 | 84.46±0.137 | 91.13±0.017 | 87.73±0.061 | 80.08±0.194 | 91.53±0.013 |
| | Co-GCN | 25.32±0.140 | 28.15±0.092 | 26.30±0.127 | 27.96±0.112 | 30.19±0.156 | 29.43±0.137 | 29.14±0.123 | 30.11±0.120 | 35.77±0.188 | 33.02±0.169 | 32.78±0.163 | 33.26±0.198 |
| | IMvGCN | 80.43±0.015 | 81.06±0.011 | 78.47±0.020 | 81.31±0.012 | 84.35±0.002 | 83.99±0.022 | 83.72±0.015 | 86.14±0.008 | 87.59±0.006 | 87.37±0.005 | 85.86±0.018 | 87.14±0.010 |
| | MMatch | 89.46±0.003 | 89.43±0.006 | 89.83±0.001 | 90.43±0.005 | 91.78±0.000 | 92.50±0.001 | 92.37±0.000 | 92.49±0.001 | 93.53±0.002 | 93.26±0.002 | 93.17±0.003 | 93.50±0.003 |
| | SPamCo | 70.02±0.006 | 61.15±0.011 | 64.43±0.018 | 87.53±0.003 | 82.48±0.083 | 80.83±0.010 | 85.66±0.019 | 81.83±0.027 | 95.15±0.010 | 94.98±0.009 | 95.42±0.005 | 94.50±0.011 |
| | TMLC | 67.85±0.001 | 67.84±0.004 | 68.06±0.000 | 68.60±0.003 | 72.37±0.000 | 72.60±0.001 | 72.39±0.001 | 72.40±0.000 | 75.14±0.000 | 74.88±0.000 | 75.39±0.000 | 74.41±0.001 |
| | CDPL | **97.34±0.001** | **97.44±0.002** | **96.73±0.002** | **98.75±0.000** | **98.19±0.001** | **98.04±0.000** | **98.05±0.000** | **98.78±0.001** | **98.28±0.002** | **98.34±0.001** | **98.26±0.001** | **98.78±0.000** |
| | Δ | **+7.88** | **+7.98** | **+6.90** | **+8.32** | **+6.41** | **+5.54** | **+5.68** | **+3.88** | **+3.13** | **+2.58** | **+2.84** | **+4.28** |
| **OutScene** | CCL | 63.94±0.004 | 63.86±0.006 | 64.18±0.011 | 60.26±0.031 | 69.55±0.022 | 70.35±0.016 | 70.19±0.005 | 64.42±0.006 | 72.28±0.007 | 73.96±0.016 | 74.36±0.005 | 70.99±0.006 |
| | CPE | 35.34±0.202 | 44.87±0.178 | 43.91±0.179 | 38.62±0.135 | 45.99±0.289 | 63.22±0.029 | 63.46±0.035 | 49.04±0.218 | 47.60±0.304 | 47.60±0.296 | 49.28±0.301 | 43.99±0.267 |
| | SimPro | 56.41±0.068 | 56.65±0.066 | 53.04±0.010 | 45.27±0.021 | 70.99±0.019 | 64.98±0.040 | 66.27±0.014 | 62.02±0.092 | 64.98±0.132 | 67.31±0.084 | 65.63±0.067 | 64.02±0.057 |
| | Meta-Expert | 70.19±0.021 | 66.75±0.010 | 68.83±0.020 | 70.59±0.041 | 74.92±0.028 | 71.71±0.019 | 73.08±0.027 | 73.72±0.019 | 73.00±0.004 | 74.52±0.011 | 71.07±0.017 | 76.68±0.017 |
| | Co-GCN | 20.31±0.014 | 20.15±0.023 | 19.71±0.023 | 19.71±0.022 | 21.35±0.014 | 21.79±0.016 | 21.51±0.018 | 20.79±0.043 | 25.84±0.046 | 28.25±0.076 | 24.64±0.039 | 23.12±0.042 |
| | DSRL | 25.00±0.000 | 22.84±0.000 | 24.28±0.000 | | 24.76±0.000 | 26.20±0.000 | 26.68±0.000 | | 25.24±0.000 | 26.44±0.000 | 25.48±0.000 | |
| | GEGCN | 67.31±0.074 | 65.14±0.038 | 66.35±0.035 | 57.61±0.022 | 72.12±0.052 | 73.88±0.016 | 74.36±0.006 | 73.24±0.005 | 72.92±0.017 | 69.47±0.035 | 69.31±0.067 | 75.72±0.017 |
| | IMvGCN | 70.67±0.013 | 68.59±0.007 | 70.67±0.035 | 55.45±0.044 | 71.15±0.029 | 66.67±0.045 | 70.35±0.027 | 61.86±0.029 | 71.15±0.017 | 71.31±0.007 | 71.96±0.023 | 68.75±0.005 |
| | MMatch | 70.11±0.012 | 70.59±0.009 | 69.71±0.015 | 64.90±0.008 | 73.88±0.001 | 72.60±0.008 | 75.88±0.016 | 72.28±0.011 | 72.04±0.005 | 72.68±0.014 | 73.88±0.005 | 76.20±0.004 |
| | SPamCo | 12.50±0.000 | 13.70±0.000 | 14.26±0.001 | 12.50±0.000 | 12.58±0.001 | 14.26±0.001 | 12.98±0.000 | 12.50±0.000 | 14.26±0.010 | 13.86±0.015 | 14.10±0.011 | 12.58±0.001 |
| | TMLC | 62.34±0.012 | 62.74±0.017 | 62.98±0.001 | 53.13±0.012 | 63.62±0.004 | 63.94±0.003 | 63.22±0.023 | 62.58±0.001 | 65.46±0.001 | 66.35±0.007 | 66.43±0.002 | 68.59±0.009 |
| | CDPL | **73.72±0.043** | **76.28±0.038** | **75.56±0.027** | **81.01±0.015** | **79.73±0.019** | **80.77±0.013** | **80.13±0.007** | **82.21±0.013** | **78.04±0.012** | **80.21±0.008** | **79.73±0.006** | **83.25±0.010** |
| | Δ | **+3.05** | **+5.69** | **+4.89** | **+10.42** | **+4.81** | **+6.89** | **+4.25** | **+8.49** | **+5.04** | **+5.69** | **+5.37** | **+6.57** |
| **YouTubeFace20** | CCL | 96.73±0.002 | 96.96±0.002 | 96.77±0.006 | 96.35±0.007 | 98.01±0.004 | 98.34±0.002 | 98.28±0.001 | 98.12±0.002 | 98.70±0.000 | 98.66±0.002 | 98.73±0.002 | 98.54±0.002 |
| | CPE | 96.76±0.004 | 97.41±0.005 | 96.02±0.022 | 97.35±0.003 | 98.45±0.002 | 98.57±0.002 | 98.62±0.002 | 98.56±0.002 | 99.00±0.002 | 96.41±0.045 | 98.89±0.001 | 98.71±0.003 |
| | SimPro | 94.70±0.008 | 92.33±0.018 | 91.50±0.027 | 91.01±0.007 | 96.09±0.008 | 96.33±0.007 | 96.29±0.011 | 92.83±0.012 | 95.83±0.005 | 96.08±0.009 | 95.63±0.003 | 93.87±0.012 |
| | Meta-Expert | 97.71±0.007 | 98.38±0.002 | 97.33±0.012 | 97.70±0.005 | 99.44±0.002 | 99.36±0.004 | 99.47±0.000 | 99.40±0.001 | 99.49±0.001 | 99.47±0.000 | 99.52±0.000 | 99.34±0.003 |
| | Co-GCN | 86.15±0.085 | 85.46±0.080 | 86.13±0.079 | 84.88±0.084 | 90.13±0.072 | 89.60±0.068 | 90.11±0.072 | 88.18±0.079 | 91.62±0.069 | 91.12±0.068 | 91.16±0.067 | 90.27±0.069 |
| | IMvGCN | 98.04±0.006 | 97.41±0.004 | 98.50±0.002 | 98.71±0.003 | 99.13±0.002 | 99.47±0.001 | 99.43±0.000 | 99.09±0.002 | 99.37±0.000 | 99.46±0.000 | 99.42±0.001 | 99.25±0.001 |
| | MMatch | 97.62±0.000 | 97.16±0.001 | 97.25±0.001 | 97.49±0.003 | 98.62±0.002 | 98.66±0.001 | 98.51±0.001 | 98.38±0.002 | 99.31±0.000 | 99.25±0.001 | 99.15±0.001 | 99.06±0.000 |
| | SPamCo | 87.89±0.049 | 84.25±0.024 | 87.60±0.043 | 92.65±0.011 | 94.18±0.013 | 91.76±0.016 | 91.87±0.003 | 93.69±0.009 | 96.02±0.008 | 95.27±0.018 | 94.74±0.016 | 95.15±0.015 |
| | TMLC | 97.29±0.000 | 97.13±0.000 | 97.14±0.000 | 97.61±0.000 | 97.93±0.000 | 97.96±0.000 | 97.95±0.000 | 98.17±0.000 | 98.87±0.000 | 98.84±0.000 | 98.89±0.000 | 98.73±0.000 |
| | CDPL | **98.60±0.002** | **99.00±0.000** | **98.83±0.000** | **98.88±0.001** | **99.54±0.000** | **99.55±0.000** | **99.51±0.000** | **99.44±0.002** | **99.51±0.000** | **99.54±0.000** | **99.55±0.000** | **99.37±0.000** |
| | Δ | **+0.56** | **+0.62** | **+0.33** | **+0.17** | **+0.10** | **+0.08** | **+0.04** | **+0.04** | **+0.02** | **+0.07** | **+0.03** | **+0.03** |

*Note:* C/I/A/U denote consistent, inverse, arbitrary, and uniform distributions. DSRL and GEGCN ran out of memory on CIFAR-10, CIFAR-100, NoisyMNIST, and YouTubeFace20. DSRL also ran out of memory for the three OutScene uniform settings; those cells are left empty. Missing runs are excluded from ranking.

TABLE II

G-MEAN (%) ON SEVEN DATASETS AT $LR = 0.09$ AND $IR \in \{0.1, 0.2, 0.3\}$ UNDER FOUR UNLABELED DISTRIBUTIONS. VALUES ARE MEAN ± REPORTED VARIATION. THE BEST AND SECOND-BEST MEANS ARE MARKED BY PINK/BOLD AND BLUE/UNDERLINED ENTRIES, RESPECTIVELY. THE Δ ROW REPORTS CDPL MINUS THE STRONGEST COMPETING METHOD (PERCENTAGE POINTS); POSITIVE/NEGATIVE DIFFERENCES ARE BOLD RED/BLUE.

| Dataset | Method | $IR = 0.1$ | | | | $IR = 0.2$ | | | | $IR = 0.3$ | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | C | I | A | U | C | I | A | U | C | I | A | U |
| **CIFAR-100** | CCL | 72.01±0.004 | 71.69±0.003 | 71.40±0.001 | 72.32±0.004 | 76.39±0.002 | 76.19±0.001 | 76.38±0.004 | 76.81±0.006 | 77.62±0.003 | 77.65±0.003 | 77.82±0.005 | 78.05±0.010 |
| | CPE | 65.32±0.007 | 63.85±0.007 | 65.92±0.002 | 66.15±0.007 | 70.36±0.002 | 70.46±0.008 | 70.89±0.007 | 69.24±0.007 | 72.88±0.006 | 72.70±0.002 | 72.75±0.014 | 73.33±0.003 |
| | SimPro | 66.80±0.010 | 68.72±0.004 | 68.70±0.013 | 63.67±0.029 | 72.65±0.011 | 73.83±0.006 | 73.46±0.008 | 69.66±0.008 | 74.84±0.010 | 75.46±0.007 | 75.52±0.007 | 72.67±0.012 |
| | Meta-Expert | 72.11±0.021 | 70.37±0.013 | 72.13±0.011 | 72.11±0.002 | 78.19±0.003 | 77.84±0.003 | 78.13±0.006 | 78.56±0.004 | 79.41±0.001 | 79.35±0.008 | 79.61±0.004 | 80.42±0.003 |
| | Co-GCN | 33.29±0.124 | 34.26±0.119 | 33.61±0.123 | 33.72±0.119 | 35.85±0.145 | 36.64±0.135 | 35.84±0.142 | 36.67±0.128 | 38.24±0.141 | 38.24±0.141 | 38.07±0.142 | 38.69±0.145 |
| | IMvGCN | 63.39±0.003 | 64.14±0.003 | 64.00±0.003 | 65.22±0.002 | 68.53±0.005 | 68.59±0.001 | 68.75±0.002 | 69.25±0.001 | 71.36±0.003 | 71.22±0.001 | 71.54±0.002 | 71.87±0.003 |
| | MMatch | 20.03±0.001 | 20.59±0.000 | 20.76±0.000 | 21.51±0.004 | 24.37±0.000 | 24.47±0.002 | 24.42±0.006 | 25.08±0.015 | 28.14±0.016 | 27.56±0.005 | 28.00±0.020 | 27.84±0.026 |
| | SPamCo | 47.21±0.147 | 56.68±0.003 | 56.35±0.002 | 57.02±0.009 | 63.52±0.003 | 63.52±0.003 | 63.52±0.003 | 64.02±0.002 | 65.85±0.003 | 65.85±0.003 | 65.85±0.003 | 67.84±0.001 |
| | TMLC | 68.45±0.000 | 68.62±0.001 | 68.67±0.000 | 68.29±0.001 | 72.36±0.000 | 72.18±0.000 | 72.43±0.001 | 72.40±0.003 | 73.02±0.001 | 73.15±0.000 | 73.01±0.000 | 74.98±0.000 |
| | CDPL | **76.62±0.002** | **76.62±0.002** | **76.62±0.002** | **77.05±0.002** | **79.74±0.002** | **79.74±0.002** | **79.74±0.002** | **79.78±0.003** | **81.10±0.004** | **81.10±0.004** | **81.10±0.004** | **81.26±0.001** |
| | Δ | **+4.51** | **+4.93** | **+4.49** | **+4.73** | **+1.55** | **+1.90** | **+1.61** | **+1.22** | **+1.69** | **+1.75** | **+1.49** | **+0.84** |
| **CIFAR-10** | CCL | 62.54±0.017 | 63.44±0.026 | 63.31±0.024 | 63.58±0.016 | 67.83±0.023 | 67.77±0.022 | 68.00±0.024 | 68.09±0.022 | 70.31±0.032 | 70.47±0.033 | 70.61±0.034 | 70.24±0.034 |
| | CPE | 58.40±0.011 | 57.19±0.016 | 57.91±0.007 | 58.62±0.004 | 61.89±0.009 | 61.51±0.003 | 60.53±0.013 | 61.50±0.009 | 63.71±0.007 | 64.24±0.009 | 62.75±0.005 | 64.60±0.012 |
| | SimPro | 71.35±0.030 | 71.61±0.020 | 71.34±0.022 | 71.38±0.023 | 73.84±0.027 | 73.28±0.023 | 73.07±0.021 | 61.68±0.117 | 74.55±0.018 | 75.03±0.022 | 74.43±0.016 | 70.22±0.069 |
| | Meta-Expert | 63.33±0.113 | 71.75±0.028 | 64.70±0.110 | 65.96±0.117 | 71.80±0.043 | 74.03±0.019 | 70.46±0.067 | 73.23±0.029 | 74.20±0.040 | 75.39±0.021 | 75.09±0.018 | 73.43±0.046 |
| | Co-GCN | 41.99±0.064 | 44.70±0.038 | 43.05±0.050 | 43.29±0.057 | 42.53±0.067 | 44.00±0.056 | 46.03±0.039 | 44.94±0.041 | 45.64±0.047 | 44.15±0.055 | 44.01±0.058 | 42.63±0.059 |
| | IMvGCN | 65.34±0.002 | 64.84±0.003 | 64.81±0.003 | 65.99±0.001 | 68.45±0.003 | 68.17±0.002 | 68.03±0.001 | 68.60±0.005 | 70.42±0.001 | 69.99±0.002 | 70.66±0.001 | 69.88±0.002 |
| | MMatch | 38.65±0.006 | 39.33±0.056 | 39.53±0.238 | 51.44±0.027 | 48.38±0.080 | 50.05±0.092 | 52.85±0.198 | 54.48±0.055 | 57.28±0.015 | 55.93±0.022 | 58.19±0.079 | 59.35±0.006 |
| | SPamCo | 58.00±0.003 | 57.52±0.002 | 57.51±0.002 | 58.41±0.000 | 63.29±0.004 | 63.29±0.004 | 63.29±0.004 | 64.09±0.001 | 67.75±0.004 | 67.59±0.003 | 67.59±0.003 | 68.06±0.003 |
| | TMLC | 71.00±0.000 | 70.86±0.000 | 70.84±0.000 | 70.22±0.000 | 72.04±0.000 | 72.06±0.000 | 72.01±0.001 | 71.87±0.000 | 73.59±0.000 | 73.40±0.000 | 73.53±0.001 | 73.26±0.000 |
| | CDPL | **71.99±0.002** | **72.90±0.000** | **72.26±0.006** | **72.59±0.001** | **75.43±0.004** | **75.29±0.002** | **75.59±0.002** | **75.40±0.003** | **76.99±0.003** | **76.73±0.002** | **76.88±0.001** | **76.73±0.003** |
| | Δ | **+0.64** | **+1.15** | **+0.92** | **+1.21** | **+1.59** | **+1.26** | **+2.52** | **+2.17** | **+2.44** | **+1.34** | **+1.79** | **+3.30** |
| **GRAZ02** | CCL | 53.00±0.016 | 51.73±0.013 | 50.45±0.005 | 60.28±0.038 | 56.50±0.007 | 54.82±0.009 | 56.83±0.012 | 61.26±0.030 | 56.72±0.004 | 55.94±0.007 | 56.94±0.005 | 65.90±0.006 |
| | CPE | 49.06±0.082 | 46.28±0.042 | 49.30±0.026 | 52.13±0.024 | 49.06±0.082 | 49.87±0.041 | 51.10±0.029 | 49.90±0.093 | 51.59±0.107 | 48.04±0.067 | 47.45±0.080 | 51.10±0.100 |
| | SimPro | 55.59±0.025 | 56.05±0.019 | 55.27±0.010 | 59.02±0.015 | 56.38±0.021 | 54.65±0.045 | 54.24±0.025 | 55.37±0.021 | 47.93±0.008 | 54.26±0.010 | 53.22±0.020 | 58.59±0.004 |
| | Meta-Expert | 54.23±0.029 | 53.57±0.011 | 53.56±0.026 | 58.92±0.002 | 57.37±0.027 | 59.34±0.018 | 57.43±0.048 | 55.09±0.047 | 55.15±0.019 | 59.89±0.007 | 60.93±0.034 | 63.60±0.018 |
| | Co-GCN | 47.03±0.031 | 46.17±0.044 | 43.68±0.035 | 45.66±0.031 | 46.02±0.035 | 48.11±0.036 | 47.94±0.032 | 47.57±0.035 | 48.73±0.033 | 47.80±0.012 | 50.72±0.022 | 48.94±0.040 |
| | DSRL | 47.08±0.000 | 45.21±0.000 | 43.30±0.000 | 52.78±0.000 | 48.90±0.000 | 49.62±0.000 | 43.30±0.000 | 51.39±0.000 | 47.81±0.000 | 43.30±0.000 | 47.45±0.000 | 54.49±0.000 |
| | GEGCN | 44.68±0.024 | 44.56±0.021 | 46.06±0.028 | 52.30±0.020 | 48.90±0.010 | 47.63±0.040 | 49.19±0.069 | 54.71±0.013 | 57.27±0.016 | 55.77±0.045 | 55.14±0.027 | 58.87±0.075 |
| | IMvGCN | 55.27±0.009 | 57.16±0.006 | 56.72±0.016 | 57.82±0.015 | 58.91±0.011 | 58.37±0.008 | 60.00±0.011 | 56.61±0.014 | 57.38±0.007 | 57.60±0.018 | 59.66±0.026 | 58.59±0.012 |
| | MMatch | 0.00±0.000 | 0.00±0.000 | 0.00±0.000 | 19.64±0.006 | 0.00±0.000 | 3.34±0.029 | 0.00±0.000 | 20.07±0.003 | 16.91±0.012 | 19.13±0.036 | 18.88±0.013 | 24.89±0.005 |
| | SPamCo | 43.30±0.000 | 43.30±0.000 | 43.30±0.000 | 43.30±0.000 | 43.56±0.002 | 43.30±0.000 | 43.30±0.000 | 43.30±0.000 | 43.30±0.000 | 43.30±0.000 | 43.30±0.000 | 43.30±0.000 |
| | TMLC | 48.91±0.009 | 50.85±0.021 | 47.38±0.009 | 48.14±0.031 | 51.92±0.002 | 51.18±0.004 | 51.81±0.018 | 48.48±0.002 | 54.03±0.003 | 53.92±0.003 | 53.50±0.015 | 54.96±0.013 |
| | CDPL | **58.91±0.019** | **63.91±0.023** | **61.92±0.014** | **62.87±0.016** | **63.39±0.024** | **63.60±0.020** | **64.44±0.018** | **62.34±0.008** | **62.87±0.016** | **63.82±0.010** | **64.86±0.006** | **66.82±0.012** |
| | Δ | **+3.32** | **+6.75** | **+5.20** | **+2.59** | **+4.48** | **+4.26** | **+4.44** | **+1.08** | **+5.49** | **+3.93** | **+3.93** | **+0.92** |
| **Handwritten** | CCL | 94.96±0.010 | 95.01±0.011 | 95.54±0.004 | 93.18±0.009 | 96.77±0.004 | 96.44±0.005 | 96.63±0.003 | 92.01±0.007 | 97.01±0.003 | 96.87±0.002 | 96.44±0.004 | 97.10±0.001 |
| | CPE | 68.56±0.108 | 74.62±0.191 | 75.49±0.190 | 60.12±0.426 | 88.83±0.084 | 87.42±0.046 | 63.72±0.398 | 75.10±0.195 | 83.33±0.096 | 89.49±0.089 | 81.92±0.192 | 72.04±0.278 |
| | SimPro | 89.49±0.008 | 90.78±0.011 | 89.68±0.019 | 87.71±0.055 | 93.79±0.027 | 95.14±0.022 | 94.86±0.023 | 94.82±0.012 | 93.72±0.043 | 93.85±0.023 | 94.33±0.020 | 93.80±0.047 |
| | Meta-Expert | 95.11±0.004 | 94.77±0.015 | 94.04±0.017 | 94.91±0.013 | 96.30±0.006 | 96.25±0.003 | 95.78±0.003 | 92.60±0.002 | 96.77±0.008 | 97.43±0.002 | 97.29±0.003 | 98.04±0.005 |
| | Co-GCN | 76.39±0.165 | 73.98±0.175 | 73.28±0.195 | 74.41±0.151 | 75.66±0.168 | 74.17±0.167 | 76.71±0.162 | 76.22±0.148 | 79.63±0.113 | 77.71±0.147 | 77.40±0.143 | 79.16±0.122 |
| | DSRL | 73.33±0.000 | 67.62±0.000 | 69.44±0.000 | 72.11±0.000 | 76.56±0.000 | 74.70±0.000 | 75.21±0.000 | 75.38±0.000 | 80.66±0.000 | 78.55±0.000 | 78.55±0.000 | 78.71±0.000 |
| | GEGCN | 86.76±0.130 | 91.90±0.051 | 78.59±0.236 | 93.09±0.042 | 77.48±0.184 | 89.91±0.083 | 95.61±0.028 | 79.61±0.185 | 87.84±0.087 | 95.85±0.032 | 94.13±0.068 | 82.69±0.071 |
| | IMvGCN | 88.64±0.010 | 87.84±0.003 | 88.94±0.014 | 89.74±0.003 | 90.44±0.008 | 89.04±0.008 | 91.91±0.014 | 92.30±0.003 | 95.01±0.002 | 93.66±0.009 | 94.72±0.002 | 95.30±0.007 |
| | MMatch | 91.24±0.010 | 89.66±0.023 | 91.47±0.025 | 92.84±0.007 | 96.43±0.007 | 95.39±0.009 | 96.44±0.000 | 97.04±0.002 | 95.92±0.001 | 95.92±0.001 | 96.05±0.006 | 96.89±0.002 |
| | SPamCo | 44.48±0.002 | 44.48±0.002 | 44.48±0.002 | 44.56±0.003 | 41.61±0.031 | 42.86±0.010 | 42.96±0.008 | 52.43±0.015 | 56.20±0.009 | 56.20±0.009 | 56.20±0.009 | 62.22±0.006 |
| | TMLC | 90.78±0.006 | 90.83±0.007 | 90.68±0.001 | 90.14±0.002 | 92.70±0.000 | 92.74±0.003 | 92.79±0.001 | 89.79±0.002 | 93.18±0.003 | 93.28±0.001 | 93.57±0.000 | 94.53±0.000 |
| | CDPL | **95.73±0.006** | **96.82±0.003** | **96.35±0.004** | **97.34±0.007** | **97.53±0.002** | **97.20±0.004** | **97.81±0.002** | **98.00±0.001** | **97.48±0.002** | **97.53±0.008** | **97.95±0.003** | **98.09±0.003** |
| | Δ | **+0.62** | **+1.81** | **+0.81** | **+2.43** | **+0.76** | **+0.76** | **+1.18** | **+0.96** | **+0.47** | **+0.10** | **+0.66** | **+0.05** |
| **NoisyMNIST** | CCL | 88.94±0.002 | 89.23±0.002 | 89.27±0.003 | 89.37±0.002 | 93.01±0.001 | 93.15±0.001 | 92.91±0.003 | 92.89±0.001 | 94.20±0.002 | 94.43±0.003 | 94.47±0.001 | 94.16±0.002 |
| | CPE | 80.32±0.011 | 81.35±0.008 | 79.55±0.009 | 81.45±0.004 | 85.06±0.001 | 87.34±0.002 | 85.86±0.010 | 85.84±0.009 | 88.14±0.003 | 88.77±0.008 | 89.29±0.020 | 88.24±0.004 |
| | SimPro | 92.09±0.030 | 94.02±0.015 | 94.13±0.004 | 94.34±0.007 | 94.10±0.002 | 95.74±0.017 | 94.12±0.005 | 97.14±0.001 | 97.00±0.012 | 97.63±0.003 | 96.83±0.008 | 96.20±0.029 |
| | Meta-Expert | 85.23±0.080 | 89.57±0.063 | 90.06±0.045 | 89.13±0.084 | 86.37±0.151 | 93.97±0.009 | 93.00±0.019 | 90.93±0.083 | 94.99±0.010 | 92.99±0.036 | 88.10±0.121 | 95.22±0.007 |
| | Co-GCN | 46.54±0.144 | 50.33±0.088 | 47.86±0.129 | 49.89±0.107 | 51.28±0.146 | 50.93±0.128 | 50.87±0.115 | 51.86±0.110 | 55.98±0.163 | 53.80±0.150 | 53.66±0.146 | 53.30±0.181 |
| | IMvGCN | 88.70±0.009 | 89.08±0.006 | 87.52±0.012 | 89.23±0.007 | 91.04±0.001 | 90.82±0.013 | 90.67±0.009 | 92.10±0.005 | 92.94±0.004 | 92.82±0.003 | 91.93±0.011 | 92.68±0.006 |
| | MMatch | 93.92±0.002 | 93.90±0.004 | 94.16±0.001 | 94.44±0.003 | 95.31±0.000 | 95.74±0.000 | 95.66±0.000 | 95.73±0.001 | 96.34±0.001 | 96.19±0.001 | 96.13±0.002 | 96.32±0.002 |
| | SPamCo | 82.27±0.004 | 76.49±0.007 | 78.67±0.012 | 92.91±0.002 | 89.87±0.050 | 88.94±0.006 | 91.81±0.011 | 89.54±0.016 | 97.28±0.006 | 97.18±0.005 | 97.43±0.003 | 96.91±0.007 |
| | TMLC | 80.88±0.000 | 80.88±0.002 | 81.02±0.000 | 81.37±0.001 | 83.76±0.000 | 83.90±0.000 | 83.76±0.000 | 83.77±0.000 | 85.48±0.000 | 85.32±0.000 | 85.63±0.000 | 85.03±0.000 |
| | CDPL | **98.51±0.000** | **98.57±0.001** | **98.17±0.001** | **99.30±0.000** | **98.99±0.000** | **98.91±0.000** | **98.91±0.000** | **99.32±0.001** | **99.04±0.001** | **99.07±0.001** | **99.03±0.001** | **99.32±0.000** |
| | Δ | **+4.59** | **+4.55** | **+4.01** | **+4.86** | **+3.68** | **+3.17** | **+3.25** | **+2.18** | **+1.76** | **+1.44** | **+1.60** | **+2.41** |
| **OutScene** | CCL | 77.88±0.003 | 77.82±0.004 | 78.04±0.007 | 75.38±0.022 | 81.56±0.014 | 82.08±0.010 | 81.98±0.003 | 78.20±0.004 | 83.31±0.005 | 84.38±0.010 | 84.64±0.003 | 82.49±0.004 |
| | CPE | 54.80±0.191 | 63.42±0.146 | 62.63±0.148 | 58.79±0.112 | 62.45±0.252 | 77.38±0.020 | 77.54±0.023 | 66.26±0.174 | 63.47±0.264 | 63.69±0.254 | 65.04±0.253 | 61.20±0.236 |
| | SimPro | 72.66±0.047 | 72.83±0.045 | 70.35±0.007 | 64.59±0.016 | 82.49±0.013 | 78.55±0.027 | 79.42±0.009 | 76.47±0.062 | 78.33±0.090 | 80.01±0.056 | 78.94±0.045 | 77.89±0.038 |
| | Meta-Expert | 81.97±0.013 | 79.73±0.007 | 81.09±0.013 | 82.22±0.027 | 84.98±0.018 | 82.95±0.013 | 83.82±0.017 | 84.23±0.012 | 83.77±0.002 | 84.74±0.007 | 82.54±0.011 | 86.10±0.011 |
| | Co-GCN | 42.41±0.016 | 42.20±0.025 | 41.72±0.026 | 41.72±0.025 | 43.52±0.015 | 43.98±0.017 | 43.67±0.019 | 42.72±0.050 | 47.93±0.043 | 49.99±0.071 | 46.77±0.039 | 45.21±0.044 |
| | DSRL | 47.25±0.000 | 45.08±0.000 | 46.53±0.000 | | 47.01±0.000 | 48.41±0.000 | 48.88±0.000 | | 47.48±0.000 | 48.65±0.000 | 47.72±0.000 | |
| | GEGCN | 80.04±0.049 | 78.66±0.025 | 79.46±0.024 | 73.56±0.015 | 83.18±0.034 | 84.33±0.010 | 84.64±0.004 | 83.93±0.003 | 83.72±0.011 | 81.50±0.023 | 81.35±0.044 | 85.49±0.011 |
| | IMvGCN | 82.29±0.008 | 80.94±0.005 | 82.27±0.022 | 72.02±0.031 | 82.59±0.019 | 79.66±0.030 | 82.07±0.017 | 76.47±0.020 | 82.59±0.011 | 82.70±0.005 | 83.10±0.015 | 81.04±0.003 |
| | MMatch | 80.46±0.007 | 80.90±0.005 | 80.23±0.008 | 76.93±0.005 | 83.70±0.001 | 82.85±0.007 | 85.19±0.010 | 82.35±0.007 | 82.46±0.003 | 83.06±0.010 | 83.86±0.005 | 85.20±0.003 |
| | SPamCo | 33.07±0.000 | 34.66±0.000 | 35.38±0.002 | 33.07±0.000 | 33.18±0.002 | 35.38±0.002 | 33.72±0.000 | 33.07±0.000 | 35.36±0.012 | 34.84±0.019 | 35.16±0.014 | 33.18±0.002 |
| | TMLC | 76.80±0.005 | 77.07±0.008 | 77.23±0.001 | 70.40±0.006 | 77.66±0.002 | 77.88±0.002 | 77.39±0.011 | 76.96±0.000 | 78.89±0.000 | 79.47±0.003 | 79.52±0.001 | 80.94±0.004 |
| | CDPL | **84.21±0.028** | **85.83±0.024** | **85.39±0.017** | **88.77±0.009** | **87.98±0.012** | **88.63±0.008** | **88.23±0.005** | **89.51±0.008** | **86.95±0.008** | **88.28±0.005** | **87.99±0.004** | **90.14±0.006** |
| | Δ | **+1.92** | **+4.89** | **+3.12** | **+6.55** | **+3.00** | **+4.30** | **+3.04** | **+5.28** | **+3.18** | **+3.54** | **+3.35** | **+4.04** |
| **YouTubeFace20** | CCL | 98.27±0.001 | 98.39±0.001 | 98.29±0.003 | 98.06±0.004 | 98.95±0.002 | 99.13±0.001 | 99.09±0.001 | 99.01±0.001 | 99.31±0.000 | 99.29±0.001 | 99.33±0.001 | 99.23±0.001 |
| | CPE | 98.28±0.002 | 98.63±0.003 | 97.88±0.012 | 98.60±0.001 | 99.18±0.001 | 99.25±0.001 | 99.27±0.001 | 99.24±0.001 | 99.47±0.001 | 98.08±0.024 | 99.42±0.001 | 99.32±0.001 |
| | SimPro | 97.18±0.004 | 95.89±0.010 | 95.43±0.015 | 95.17±0.004 | 97.92±0.004 | 98.05±0.004 | 98.03±0.006 | 96.16±0.007 | 97.78±0.003 | 97.92±0.005 | 97.68±0.002 | 96.73±0.006 |
| | Meta-Expert | 98.79±0.004 | 99.15±0.001 | 98.59±0.006 | 98.79±0.003 | 99.71±0.001 | 99.54±0.004 | 99.72±0.000 | 99.68±0.000 | 99.73±0.000 | 99.72±0.000 | 99.75±0.000 | 99.50±0.004 |
| | Co-GCN | 92.39±0.048 | 92.01±0.045 | 92.39±0.044 | 91.68±0.047 | 94.63±0.039 | 94.35±0.038 | 94.62±0.040 | 93.54±0.044 | 95.46±0.038 | 95.18±0.037 | 95.21±0.037 | 94.71±0.038 |
| | IMvGCN | 98.96±0.003 | 98.63±0.002 | 99.21±0.001 | 99.32±0.001 | 99.54±0.001 | 99.72±0.000 | 99.70±0.000 | 99.52±0.001 | 99.67±0.000 | 99.72±0.000 | 99.69±0.000 | 99.61±0.000 |
| | MMatch | 98.72±0.000 | 98.47±0.001 | 98.52±0.001 | 98.66±0.001 | 99.26±0.001 | 99.28±0.000 | 99.20±0.001 | 99.14±0.001 | 99.63±0.000 | 99.60±0.001 | 99.55±0.001 | 99.50±0.000 |
| | SPamCo | 93.43±0.028 | 91.40±0.013 | 93.27±0.024 | 96.07±0.006 | 96.90±0.007 | 95.58±0.009 | 95.64±0.002 | 96.63±0.005 | 97.89±0.004 | 97.48±0.010 | 97.20±0.009 | 97.42±0.008 |
| | TMLC | 98.56±0.000 | 98.48±0.000 | 98.49±0.000 | 98.73±0.000 | 98.91±0.000 | 98.92±0.000 | 98.92±0.000 | 99.03±0.000 | 99.40±0.000 | 99.39±0.000 | 99.41±0.000 | 99.33±0.000 |
| | CDPL | **99.26±0.001** | **99.48±0.000** | **99.38±0.000** | **99.41±0.001** | **99.76±0.000** | **99.76±0.000** | **99.74±0.000** | **99.70±0.001** | **99.74±0.000** | **99.76±0.000** | **99.76±0.000** | **99.67±0.000** |
| | Δ | **+0.30** | **+0.33** | **+0.17** | **+0.09** | **+0.05** | **+0.04** | **+0.02** | **+0.02** | **+0.01** | **+0.04** | **+0.01** | **+0.06** |

*Note:* C/I/A/U denote consistent, inverse, arbitrary, and uniform distributions. DSRL and GEGCN ran out of memory on CIFAR-10, CIFAR-100, NoisyMNIST, and YouTubeFace20. DSRL also ran out of memory for the three OutScene uniform settings; those cells are left empty. Missing runs are excluded from ranking.

TABLE III

SPECIFICITY (%) ON SEVEN DATASETS AT $LR = 0.09$ AND $IR \in \{0.1, 0.2, 0.3\}$ UNDER FOUR UNLABELED DISTRIBUTIONS. VALUES ARE MEAN ± REPORTED VARIATION. THE BEST AND SECOND-BEST MEANS ARE MARKED BY PINK/BOLD AND BLUE/UNDERLINED ENTRIES, RESPECTIVELY. THE Δ ROW REPORTS CDPL MINUS THE STRONGEST COMPETING METHOD (PERCENTAGE POINTS); POSITIVE/NEGATIVE DIFFERENCES ARE BOLD RED/BLUE.

| Dataset | Method | $IR = 0.1$ | | | | $IR = 0.2$ | | | | $IR = 0.3$ | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | C | I | A | U | C | I | A | U | C | I | A | U |
| **CIFAR-100** | CCL | 99.52±0.000 | 99.51±0.000 | 99.51±0.000 | 99.52±0.000 | 99.58±0.000 | 99.58±0.000 | 99.58±0.000 | 99.59±0.000 | 99.60±0.000 | 99.60±0.000 | 99.60±0.000 | 99.61±0.000 |
| | CPE | 99.42±0.000 | 99.40±0.000 | 99.43±0.000 | 99.43±0.000 | 99.49±0.000 | 99.49±0.000 | 99.50±0.000 | 99.48±0.000 | 99.53±0.000 | 99.53±0.000 | 99.53±0.000 | 99.54±0.000 |
| | SimPro | 99.44±0.000 | 99.47±0.000 | 99.47±0.000 | 99.40±0.000 | 99.53±0.000 | 99.54±0.000 | 99.54±0.000 | 99.48±0.000 | 99.56±0.000 | 99.57±0.000 | 99.57±0.000 | 99.53±0.000 |
| | Meta-Expert | 99.52±0.000 | 99.49±0.000 | 99.52±0.000 | 99.52±0.000 | 99.61±0.000 | 99.60±0.000 | 99.61±0.000 | 99.62±0.000 | 99.63±0.000 | 99.63±0.000 | 99.63±0.000 | 99.65±0.000 |
| | Co-GCN | 99.12±0.001 | 99.12±0.001 | 99.12±0.001 | 99.12±0.001 | 99.14±0.001 | 99.14±0.001 | 99.14±0.001 | 99.14±0.001 | 99.16±0.001 | 99.16±0.001 | 99.15±0.001 | 99.16±0.001 |
| | IMvGCN | 99.40±0.000 | 99.41±0.000 | 99.41±0.000 | 99.42±0.000 | 99.47±0.000 | 99.47±0.000 | 99.47±0.000 | 99.48±0.000 | 99.51±0.000 | 99.50±0.000 | 99.51±0.000 | 99.51±0.000 |
| | MMatch | 99.13±0.000 | 99.13±0.000 | 99.13±0.000 | 99.13±0.000 | 99.15±0.000 | 99.15±0.000 | 99.15±0.000 | 99.15±0.000 | 99.16±0.000 | 99.16±0.000 | 99.16±0.000 | 99.16±0.000 |
| | SPamCo | 99.23±0.001 | 99.32±0.000 | 99.31±0.000 | 99.32±0.000 | 99.40±0.000 | 99.40±0.000 | 99.40±0.000 | 99.41±0.000 | 99.43±0.000 | 99.43±0.000 | 99.43±0.000 | 99.46±0.000 |
| | TMLC | 99.47±0.000 | 99.47±0.000 | 99.47±0.000 | 99.46±0.000 | 99.52±0.000 | 99.52±0.000 | 99.52±0.000 | 99.52±0.000 | 99.53±0.000 | 99.53±0.000 | 99.53±0.000 | 99.56±0.000 |
| | CDPL | **99.59±0.000** | **99.59±0.000** | **99.59±0.000** | **99.59±0.000** | **99.63±0.000** | **99.63±0.000** | **99.63±0.000** | **99.64±0.000** | **99.66±0.000** | **99.66±0.000** | **99.66±0.000** | **99.66±0.000** |
| | Δ | **+0.07** | **+0.08** | **+0.07** | **+0.07** | **+0.02** | **+0.03** | **+0.02** | **+0.02** | **+0.03** | **+0.03** | **+0.03** | **+0.01** |
| **CIFAR-10** | CCL | 93.54±0.002 | 93.67±0.004 | 93.65±0.003 | 93.69±0.002 | 94.31±0.003 | 94.30±0.003 | 94.34±0.004 | 94.35±0.003 | 94.70±0.005 | 94.72±0.005 | 94.74±0.005 | 94.69±0.005 |
| | CPE | 92.97±0.002 | 92.81±0.002 | 92.90±0.001 | 93.00±0.001 | 93.44±0.001 | 93.39±0.001 | 93.26±0.002 | 93.39±0.001 | 93.70±0.001 | 93.78±0.001 | 93.56±0.001 | 93.83±0.002 |
| | SimPro | 94.86±0.005 | 94.90±0.003 | 94.85±0.004 | 94.86±0.004 | 95.25±0.004 | 95.16±0.004 | 95.13±0.003 | 93.50±0.016 | 95.37±0.003 | 95.44±0.004 | 95.35±0.003 | 94.70±0.011 |
| | Meta-Expert | 93.73±0.015 | 94.92±0.004 | 93.92±0.015 | 94.11±0.017 | 94.93±0.007 | 95.28±0.003 | 94.74±0.010 | 95.16±0.005 | 95.32±0.006 | 95.50±0.003 | 95.45±0.003 | 95.20±0.007 |
| | Co-GCN | 91.08±0.006 | 91.33±0.004 | 91.17±0.005 | 91.20±0.006 | 91.14±0.007 | 91.27±0.006 | 91.48±0.004 | 91.36±0.004 | 91.44±0.005 | 91.29±0.006 | 91.28±0.006 | 91.14±0.006 |
| | IMvGCN | 93.94±0.000 | 93.87±0.000 | 93.86±0.000 | 94.03±0.000 | 94.40±0.001 | 94.36±0.000 | 94.34±0.000 | 94.43±0.001 | 94.71±0.000 | 94.64±0.000 | 94.74±0.000 | 94.62±0.000 |
| | MMatch | 92.12±0.000 | 92.21±0.000 | 92.16±0.001 | 92.96±0.000 | 92.75±0.000 | 92.80±0.000 | 93.05±0.001 | 93.14±0.001 | 93.43±0.000 | 93.26±0.000 | 93.48±0.001 | 93.58±0.000 |
| | SPamCo | 92.91±0.000 | 92.85±0.000 | 92.85±0.000 | 92.97±0.000 | 93.64±0.001 | 93.64±0.001 | 93.64±0.001 | 93.76±0.000 | 94.30±0.001 | 94.27±0.001 | 94.27±0.001 | 94.35±0.000 |
| | TMLC | 94.80±0.000 | 94.78±0.000 | 94.77±0.000 | 94.68±0.000 | 94.96±0.000 | 94.96±0.000 | 94.96±0.000 | 94.93±0.000 | 95.21±0.000 | 95.18±0.000 | 95.20±0.000 | 95.16±0.000 |
| | CDPL | **94.95±0.000** | **95.10±0.000** | **95.00±0.001** | **95.05±0.000** | **95.51±0.001** | **95.49±0.000** | **95.53±0.000** | **95.50±0.001** | **95.77±0.001** | **95.72±0.000** | **95.75±0.000** | **95.72±0.001** |
| | Δ | **+0.09** | **+0.18** | **+0.15** | **+0.19** | **+0.26** | **+0.21** | **+0.40** | **+0.34** | **+0.40** | **+0.22** | **+0.30** | **+0.52** |
| **GRAZ02** | CCL | 78.58±0.006 | 78.09±0.005 | 77.60±0.002 | 81.54±0.016 | 79.97±0.003 | 79.30±0.004 | 80.11±0.005 | 81.94±0.013 | 80.06±0.002 | 79.75±0.003 | 80.15±0.002 | 83.92±0.003 |
| | CPE | 77.15±0.030 | 76.08±0.015 | 77.17±0.010 | 78.25±0.009 | 77.15±0.030 | 77.40±0.016 | 77.86±0.011 | 77.49±0.035 | 78.15±0.041 | 76.75±0.025 | 76.55±0.029 | 77.95±0.038 |
| | SimPro | 79.61±0.010 | 79.79±0.008 | 79.48±0.004 | 81.00±0.006 | 79.93±0.009 | 79.26±0.018 | 79.08±0.010 | 79.53±0.009 | 76.66±0.003 | 79.08±0.004 | 78.67±0.008 | 80.82±0.002 |
| | Meta-Expert | 79.08±0.011 | 78.81±0.004 | 78.81±0.010 | 80.96±0.001 | 80.33±0.011 | 81.14±0.008 | 80.38±0.019 | 79.44±0.019 | 79.44±0.008 | 81.36±0.003 | 81.81±0.014 | 82.93±0.008 |
| | Co-GCN | 76.34±0.011 | 76.05±0.016 | 75.16±0.012 | 75.85±0.011 | 75.99±0.013 | 76.75±0.013 | 76.68±0.012 | 76.55±0.013 | 76.97±0.012 | 76.61±0.005 | 77.71±0.008 | 77.06±0.015 |
| | DSRL | 76.34±0.000 | 75.67±0.000 | 75.00±0.000 | 78.49±0.000 | 77.02±0.000 | 77.28±0.000 | 75.00±0.000 | 77.96±0.000 | 76.61±0.000 | 75.00±0.000 | 76.48±0.000 | 79.17±0.000 |
| | GEGCN | 75.49±0.009 | 75.45±0.007 | 75.99±0.010 | 78.32±0.008 | 77.02±0.004 | 76.57±0.014 | 77.20±0.025 | 79.26±0.005 | 80.29±0.006 | 79.70±0.018 | 79.44±0.011 | 81.00±0.031 |
| | IMvGCN | 79.48±0.003 | 80.24±0.002 | 80.06±0.006 | 80.51±0.006 | 80.96±0.005 | 80.73±0.003 | 81.41±0.005 | 80.02±0.005 | 80.33±0.003 | 80.42±0.007 | 81.27±0.011 | 80.82±0.005 |
| | MMatch | 74.96±0.001 | 75.00±0.000 | 75.00±0.000 | 76.34±0.000 | 75.00±0.000 | 75.09±0.001 | 74.87±0.000 | 76.70±0.001 | 75.67±0.001 | 76.12±0.004 | 76.12±0.003 | 77.28±0.001 |
| | SPamCo | 75.00±0.000 | 75.00±0.000 | 75.00±0.000 | 75.00±0.000 | 75.09±0.001 | 75.00±0.000 | 75.00±0.000 | 75.00±0.000 | 75.00±0.000 | 75.00±0.000 | 75.00±0.000 | 75.00±0.000 |
| | TMLC | **80.96±0.001** | 78.77±0.004 | 77.33±0.001 | 77.65±0.005 | 79.21±0.000 | 78.90±0.001 | 79.17±0.003 | 77.78±0.000 | 80.11±0.001 | 80.06±0.000 | 79.89±0.003 | 80.51±0.002 |
| | CDPL | **80.96±0.008** | **83.06±0.010** | **82.21±0.006** | **82.62±0.007** | **82.84±0.010** | **82.93±0.008** | **83.29±0.008** | **82.39±0.004** | **82.62±0.007** | **83.02±0.004** | **83.47±0.003** | **84.32±0.005** |
| | Δ | **0.00** | **+2.82** | **+2.15** | **+1.08** | **+1.88** | **+1.79** | **+1.88** | **+0.45** | **+2.29** | **+1.66** | **+1.66** | **+0.40** |
| **Handwritten** | CCL | 99.01±0.002 | 99.02±0.002 | 99.12±0.001 | 98.67±0.002 | 99.36±0.001 | 99.30±0.001 | 99.33±0.001 | 98.44±0.001 | 99.41±0.001 | 99.38±0.000 | 99.30±0.001 | 99.43±0.000 |
| | CPE | 94.47±0.017 | 95.53±0.031 | 95.67±0.031 | 94.06±0.057 | 97.88±0.016 | 97.60±0.008 | 94.43±0.056 | 95.61±0.032 | 96.89±0.017 | 98.00±0.017 | 96.75±0.033 | 95.29±0.044 |
| | SimPro | 97.97±0.001 | 98.21±0.002 | 98.01±0.004 | 97.66±0.010 | 98.79±0.005 | 99.05±0.004 | 98.99±0.004 | 98.98±0.002 | 98.78±0.008 | 98.80±0.004 | 98.89±0.004 | 98.80±0.009 |
| | Meta-Expert | 99.04±0.001 | 98.97±0.003 | 98.83±0.003 | 99.00±0.003 | 99.27±0.001 | 99.26±0.001 | 99.17±0.001 | 98.56±0.000 | 99.36±0.002 | 99.49±0.000 | 99.46±0.001 | 99.61±0.001 |
| | Co-GCN | 95.85±0.027 | 95.49±0.028 | 95.43±0.031 | 95.50±0.024 | 95.74±0.027 | 95.50±0.027 | 95.90±0.027 | 95.79±0.024 | 96.30±0.019 | 96.03±0.025 | 95.97±0.024 | 96.23±0.021 |
| | DSRL | 95.17±0.000 | 94.28±0.000 | 94.56±0.000 | 94.97±0.000 | 95.69±0.000 | 95.39±0.000 | 95.47±0.000 | 95.50±0.000 | 96.39±0.000 | 96.03±0.000 | 96.03±0.000 | 96.06±0.000 |
| | GEGCN | 97.56±0.023 | 98.44±0.010 | 96.34±0.038 | 98.66±0.008 | 96.03±0.031 | 98.08±0.015 | 99.14±0.006 | 96.39±0.031 | 97.70±0.016 | 99.19±0.006 | 98.87±0.013 | 96.77±0.012 |
| | IMvGCN | 97.81±0.002 | 97.67±0.001 | 97.87±0.003 | 98.02±0.001 | 98.15±0.001 | 97.89±0.002 | 98.43±0.003 | 98.50±0.001 | 99.02±0.000 | 98.76±0.002 | 98.96±0.000 | 99.07±0.001 |
| | MMatch | 98.46±0.001 | 98.31±0.003 | 98.53±0.004 | 98.74±0.001 | 99.31±0.001 | 99.11±0.002 | 99.31±0.000 | 99.42±0.000 | 99.21±0.000 | 99.21±0.000 | 99.24±0.001 | 99.39±0.000 |
| | SPamCo | 91.30±0.000 | 91.30±0.000 | 91.30±0.000 | 91.31±0.000 | 91.01±0.003 | 91.13±0.001 | 91.14±0.001 | 92.20±0.002 | 92.68±0.001 | 92.68±0.001 | 92.68±0.001 | 93.49±0.001 |
| | TMLC | 98.21±0.000 | 98.22±0.000 | 98.19±0.000 | 98.09±0.000 | 98.57±0.000 | 98.58±0.000 | 98.59±0.000 | 98.03±0.000 | 98.67±0.000 | 98.69±0.000 | 98.74±0.000 | 98.93±0.000 |
| | CDPL | **99.16±0.001** | **99.37±0.001** | **99.28±0.001** | **99.47±0.001** | **99.51±0.000** | **99.44±0.001** | **99.56±0.000** | **99.60±0.000** | **99.50±0.001** | **99.51±0.002** | **99.59±0.001** | **99.62±0.001** |
| | Δ | **+0.12** | **+0.35** | **+0.16** | **+0.47** | **+0.15** | **+0.14** | **+0.23** | **+0.18** | **+0.09** | **+0.02** | **+0.13** | **+0.01** |
| **NoisyMNIST** | CCL | 97.87±0.000 | 97.92±0.000 | 97.93±0.001 | 97.95±0.001 | 98.63±0.000 | 98.66±0.000 | 98.62±0.001 | 98.61±0.000 | 98.86±0.000 | 98.91±0.001 | 98.91±0.000 | 98.85±0.000 |
| | CPE | 96.33±0.002 | 96.51±0.001 | 96.20±0.002 | 96.53±0.001 | 97.16±0.000 | 97.58±0.000 | 97.31±0.002 | 97.30±0.002 | 97.72±0.001 | 97.84±0.002 | 97.94±0.004 | 97.74±0.001 |
| | SimPro | 98.46±0.006 | 98.83±0.003 | 98.85±0.001 | 98.89±0.001 | 98.84±0.000 | 99.16±0.003 | 98.85±0.001 | 99.43±0.000 | 99.41±0.002 | 99.53±0.001 | 99.37±0.002 | 99.25±0.006 |
| | Meta-Expert | 97.23±0.014 | 98.00±0.012 | 98.09±0.008 | 97.94±0.015 | 97.51±0.026 | 98.82±0.002 | 98.63±0.004 | 98.27±0.015 | 99.01±0.002 | 98.64±0.007 | 97.79±0.022 | 99.06±0.001 |
| | Co-GCN | 91.70±0.016 | 92.02±0.010 | 91.81±0.014 | 92.00±0.013 | 92.24±0.017 | 92.16±0.015 | 92.13±0.014 | 92.23±0.013 | 92.86±0.021 | 92.56±0.019 | 92.53±0.018 | 92.58±0.022 |
| | IMvGCN | 97.83±0.002 | 97.90±0.001 | 97.61±0.002 | 97.92±0.001 | 98.26±0.000 | 98.22±0.002 | 98.19±0.002 | 98.46±0.001 | 98.62±0.001 | 98.60±0.001 | 98.43±0.002 | 98.57±0.001 |
| | MMatch | 98.83±0.000 | 98.83±0.001 | 98.87±0.000 | 98.94±0.001 | 99.09±0.000 | 99.17±0.000 | 99.15±0.000 | 99.17±0.000 | 99.28±0.000 | 99.25±0.000 | 99.24±0.000 | 99.28±0.000 |
| | SPamCo | 96.67±0.001 | 95.68±0.001 | 96.05±0.002 | 98.61±0.000 | 98.05±0.009 | 97.87±0.001 | 98.41±0.002 | 97.98±0.003 | 99.46±0.001 | 99.44±0.001 | 99.49±0.001 | 99.39±0.001 |
| | TMLC | 96.43±0.000 | 96.43±0.000 | 96.45±0.000 | 96.51±0.000 | 96.93±0.000 | 96.96±0.000 | 96.93±0.000 | 96.93±0.000 | 97.24±0.000 | 97.21±0.000 | 97.27±0.000 | 97.16±0.000 |
| | CDPL | **99.70±0.000** | **99.72±0.000** | **99.64±0.000** | **99.86±0.000** | **99.80±0.000** | **99.78±0.000** | **99.78±0.000** | **99.86±0.000** | **99.81±0.000** | **99.82±0.000** | **99.81±0.000** | **99.86±0.000** |
| | Δ | **+0.87** | **+0.89** | **+0.77** | **+0.92** | **+0.71** | **+0.61** | **+0.63** | **+0.43** | **+0.35** | **+0.29** | **+0.32** | **+0.47** |
| **OutScene** | CCL | 94.85±0.001 | 94.84±0.001 | 94.88±0.002 | 94.32±0.005 | 95.65±0.003 | 95.76±0.002 | 95.74±0.001 | 94.92±0.001 | 96.04±0.001 | 96.28±0.002 | 96.34±0.001 | 95.86±0.001 |
| | CPE | 90.76±0.029 | 92.12±0.026 | 91.99±0.026 | 91.23±0.019 | 92.28±0.041 | 94.75±0.004 | 94.78±0.005 | 92.72±0.031 | 92.51±0.044 | 92.51±0.042 | 92.75±0.043 | 92.00±0.038 |
| | SimPro | 93.77±0.010 | 93.81±0.009 | 93.29±0.001 | 92.18±0.003 | 95.86±0.003 | 95.00±0.006 | 95.18±0.002 | 94.57±0.013 | 95.00±0.019 | 95.33±0.012 | 95.09±0.010 | 94.86±0.008 |
| | Meta-Expert | 95.74±0.003 | 95.25±0.001 | 95.55±0.003 | 95.80±0.006 | 96.42±0.004 | 95.96±0.003 | 96.15±0.004 | 96.25±0.003 | 96.14±0.001 | 96.36±0.002 | 95.87±0.002 | 96.67±0.002 |
| | Co-GCN | 88.62±0.002 | 88.59±0.003 | 88.53±0.003 | 88.53±0.003 | 88.76±0.002 | 88.83±0.002 | 88.79±0.003 | 88.68±0.006 | 89.41±0.007 | 89.75±0.011 | 89.23±0.006 | 89.02±0.006 |
| | DSRL | 89.29±0.000 | 88.98±0.000 | 89.18±0.000 | | 89.25±0.000 | 89.46±0.000 | 89.53±0.000 | | 89.32±0.000 | 89.49±0.000 | 89.35±0.000 | |
| | GEGCN | 95.33±0.011 | 95.02±0.005 | 95.19±0.005 | 93.94±0.003 | 96.02±0.007 | 96.27±0.002 | 96.34±0.001 | 96.18±0.001 | 96.13±0.003 | 95.64±0.005 | 95.62±0.010 | 96.53±0.003 |
| | IMvGCN | 95.81±0.002 | 95.51±0.001 | 95.81±0.005 | 93.64±0.006 | 95.88±0.004 | 95.24±0.007 | 95.76±0.004 | 94.55±0.004 | 95.88±0.003 | 95.90±0.001 | 95.99±0.003 | 95.54±0.001 |
| | MMatch | 95.73±0.002 | 95.80±0.001 | 95.67±0.002 | 94.99±0.001 | 96.27±0.000 | 96.09±0.001 | 96.55±0.002 | 96.04±0.002 | 96.01±0.001 | 96.10±0.002 | 96.27±0.001 | 96.60±0.001 |
| | SPamCo | 87.50±0.000 | 87.67±0.000 | 87.75±0.000 | 87.50±0.000 | 87.51±0.000 | 87.75±0.000 | 87.57±0.000 | 87.50±0.000 | 87.75±0.001 | 87.69±0.002 | 87.73±0.002 | 87.51±0.000 |
| | TMLC | 94.62±0.000 | 94.68±0.000 | 94.71±0.000 | 93.30±0.000 | 94.80±0.000 | 94.85±0.000 | 94.75±0.000 | 94.65±0.000 | 95.07±0.000 | 95.19±0.000 | 95.20±0.000 | 95.51±0.000 |
| | CDPL | **96.25±0.006** | **96.61±0.005** | **96.51±0.004** | **97.29±0.002** | **97.10±0.003** | **97.25±0.002** | **97.16±0.001** | **97.46±0.002** | **96.86±0.002** | **97.17±0.001** | **97.10±0.001** | **97.61±0.001** |
| | Δ | **+0.44** | **+0.81** | **+0.70** | **+1.49** | **+0.68** | **+0.98** | **+0.61** | **+1.21** | **+0.72** | **+0.81** | **+0.76** | **+0.94** |
| **YouTubeFace20** | CCL | 99.83±0.000 | 99.84±0.000 | 99.83±0.000 | 99.81±0.000 | 99.90±0.000 | 99.91±0.000 | 99.91±0.000 | 99.90±0.000 | 99.93±0.000 | 99.93±0.000 | 99.93±0.000 | 99.92±0.000 |
| | CPE | 99.83±0.000 | 99.86±0.000 | 99.79±0.001 | 99.86±0.000 | 99.92±0.000 | 99.92±0.000 | 99.93±0.000 | 99.92±0.000 | 99.95±0.000 | 99.81±0.002 | 99.94±0.000 | 99.93±0.000 |
| | SimPro | 99.72±0.000 | 99.60±0.001 | 99.55±0.001 | 99.53±0.000 | 99.79±0.000 | 99.81±0.000 | 99.80±0.001 | 99.62±0.001 | 99.78±0.000 | 99.79±0.001 | 99.77±0.000 | 99.68±0.001 |
| | Meta-Expert | 99.88±0.000 | 99.91±0.000 | 99.86±0.001 | 99.88±0.000 | 99.97±0.000 | 99.68±0.005 | **99.97±0.000** | **99.97±0.000** | **99.97±0.000** | 99.97±0.000 | 99.97±0.000 | 99.67±0.005 |
| | Co-GCN | 99.27±0.005 | 99.23±0.004 | 99.27±0.004 | 99.20±0.004 | 99.48±0.004 | 99.45±0.004 | 99.48±0.004 | 99.38±0.004 | 99.56±0.004 | 99.53±0.004 | 99.53±0.004 | 99.49±0.004 |
| | IMvGCN | 99.90±0.000 | 99.86±0.000 | 99.92±0.000 | 99.93±0.000 | 99.95±0.000 | 99.97±0.000 | **99.97±0.000** | 99.95±0.000 | **99.97±0.000** | 99.97±0.000 | 99.97±0.000 | 99.96±0.000 |
| | MMatch | 99.87±0.000 | 99.85±0.000 | 99.86±0.000 | 99.87±0.000 | 99.93±0.000 | 99.93±0.000 | 99.92±0.000 | 99.91±0.000 | 99.96±0.000 | 99.96±0.000 | 99.96±0.000 | 99.95±0.000 |
| | SPamCo | 99.36±0.003 | 99.17±0.001 | 99.35±0.002 | 99.61±0.001 | 99.69±0.001 | 99.57±0.001 | 99.57±0.000 | 99.67±0.001 | 99.79±0.000 | 99.75±0.001 | 99.72±0.001 | 99.74±0.001 |
| | TMLC | 99.86±0.000 | 99.85±0.000 | 99.85±0.000 | 99.87±0.000 | 99.89±0.000 | 99.89±0.000 | 99.89±0.000 | 99.90±0.000 | 99.94±0.000 | 99.94±0.000 | 99.94±0.000 | 99.93±0.000 |
| | CDPL | **99.93±0.000** | **99.95±0.000** | **99.94±0.000** | **99.94±0.000** | **99.98±0.000** | **99.98±0.000** | **99.97±0.000** | **99.97±0.000** | **99.97±0.000** | **99.98±0.000** | **99.98±0.000** | **99.97±0.000** |
| | Δ | **+0.03** | **+0.04** | **+0.02** | **+0.01** | **+0.01** | **+0.01** | **0.00** | **0.00** | **0.00** | **+0.01** | **+0.01** | **+0.01** |

*Note:* C/I/A/U denote consistent, inverse, arbitrary, and uniform distributions. DSRL and GEGCN ran out of memory on CIFAR-10, CIFAR-100, NoisyMNIST, and YouTubeFace20. DSRL also ran out of memory for the three OutScene uniform settings; those cells are left empty. Missing runs are excluded from ranking.

TABLE IV

F1 SCORES (%) UNDER THE CONSISTENT LONG-TAIL SETTING ($IR = 0.1$) ACROSS LABELING RATIOS $LR \in \{0.03, 0.06, 0.09, 0.12, 0.15\}$. RESULTS ARE MEAN ± STANDARD DEVIATION OVER THREE SEEDS; "—" INDICATES OUT-OF-MEMORY RUNS. BEST AND SECOND-BEST RESULTS ARE MARKED IN PINK/BOLD AND BLUE/UNDERLINED, RESPECTIVELY. Δ DENOTES CDPL MINUS THE BEST COMPETITOR (PERCENTAGE POINTS), WITH POSITIVE/NEGATIVE VALUES IN BOLD RED/BLUE.

| Dataset | Method | LR=0.03 | LR=0.06 | LR=0.09 | LR=0.12 | LR=0.15 |
|---|---|---|---|---|---|---|
| CIFAR100 | CCL | 30.59 | 45.12 | 50.46 | 47.66 | 48.08 |
| | CPE | 21.35 | 35.71 | 40.35 | 38.73 | 38.72 |
| | SimPro | 27.09 | 37.63 | 44.58 | 45.42 | 44.39 |
| | Meta-Expert | 24.44 | 42.84 | 53.38 | 50.79 | 50.12 |
| | Co-GCN | 12.62 | 14.16 | 8.12 | 14.72 | 14.53 |
| | DSRL | - | - | - | - | - |
| | GEGCN | - | - | - | - | - |
| | IMvGCN | 28.88 | 35.19 | 36.82 | 39.58 | 40.42 |
| | MMatch | 8.78 | 8.14 | 7.00 | 7.33 | 6.98 |
| | SPamCo | 13.73 | 20.40 | 20.42 | 32.06 | 36.84 |
| | TMLC | 34.32 | 38.57 | 39.43 | 40.72 | 41.25 |
| | Ours | **42.31** | **53.86** | **58.27** | **62.20** | **64.29** |
| | Δ | **+7.99** | **+8.74** | **+4.89** | **+11.41** | **+14.17** |
| GRAZ02 | CCL | 34.81 | 33.14 | 32.15 | 28.63 | 30.43 |
| | CPE | 31.74 | 23.19 | 21.65 | 27.03 | 27.75 |
| | SimPro | 35.96 | 27.34 | 36.08 | 25.79 | 37.54 |
| | Meta-Expert | 23.12 | 32.15 | 36.20 | 32.63 | 35.15 |
| | Co-GCN | 13.91 | 15.23 | 18.36 | 12.73 | 16.68 |
| | DSRL | 20.70 | 10.00 | 23.21 | 10.00 | 10.00 |
| | GEGCN | 17.14 | 21.91 | 13.40 | 20.69 | 28.10 |
| | IMvGCN | 34.26 | 29.52 | 38.13 | 32.89 | 40.46 |
| | MMatch | 12.12 | 10.00 | 9.96 | 10.83 | 13.67 |
| | SPamCo | 10.00 | 10.00 | 10.00 | 10.55 | 10.00 |
| | TMLC | 34.41 | 30.99 | 33.04 | 26.44 | 38.10 |
| | Ours | **39.80** | **40.01** | **40.89** | **39.12** | **49.12** |
| | Δ | **+3.84** | **+6.87** | **+2.76** | **+6.23** | **+8.66** |
| NoisyMNIST | CCL | 68.22 | 77.05 | 80.23 | 80.69 | 81.53 |
| | CPE | 46.61 | 60.33 | 65.32 | 65.79 | 61.51 |
| | SimPro | 79.02 | 82.87 | 86.20 | 90.41 | 84.96 |
| | Meta-Expert | 77.31 | 81.50 | 75.75 | 86.47 | 84.94 |
| | Co-GCN | 22.86 | 22.58 | 14.06 | 22.93 | 23.29 |
| | IMvGCN | 72.69 | 77.64 | 79.59 | 80.35 | 79.31 |
| | MMatch | 81.54 | 88.02 | 89.35 | 90.67 | 90.83 |
| | SPamCo | 26.10 | 40.88 | 63.02 | 64.83 | 80.69 |
| | TMLC | 61.24 | 65.27 | 65.75 | 67.47 | 69.03 |
| | Ours | **96.95** | **97.44** | **97.34** | **97.45** | **97.63** |
| | Δ | **+15.41** | **+9.42** | **+7.99** | **+6.78** | **+6.80** |
| OutScene | CCL | 49.57 | 64.18 | 62.13 | 59.18 | 55.79 |
| | CPE | 36.80 | 43.63 | 29.91 | 45.17 | 41.77 |
| | SimPro | 41.85 | 48.47 | 55.11 | 52.82 | 48.76 |
| | Meta-Expert | 50.79 | 66.73 | 67.66 | 66.90 | 59.54 |
| | Co-GCN | 14.49 | 15.79 | 13.00 | 13.11 | 12.66 |
| | DSRL | 22.53 | 25.22 | 23.18 | 24.86 | 21.96 |
| | GEGCN | 36.71 | 48.67 | 62.88 | 73.91 | 67.99 |
| | IMvGCN | 57.84 | 66.00 | 69.37 | 69.28 | 64.83 |
| | MMatch | 53.53 | 63.32 | 68.44 | 69.34 | 70.56 |
| | SPamCo | 2.78 | 2.78 | 2.78 | 2.78 | 5.58 |
| | TMLC | 44.38 | 54.28 | 60.88 | 63.52 | 58.44 |
| | Ours | **60.95** | **68.24** | **73.36** | **76.99** | **74.39** |
| | Δ | **+3.11** | **+1.51** | **+3.99** | **+3.08** | **+3.83** |

TABLE V

ACCURACY (%) UNDER THE CONSISTENT LONG-TAIL SETTING ($IR = 0.1$) ACROSS LABELING RATIOS $LR \in \{0.03, 0.06, 0.09, 0.12, 0.15\}$. RESULTS ARE MEAN ± STANDARD DEVIATION OVER THREE SEEDS; "—" INDICATES OUT-OF-MEMORY RUNS. BEST AND SECOND-BEST RESULTS ARE MARKED IN PINK/BOLD AND BLUE/UNDERLINED, RESPECTIVELY. Δ DENOTES CDPL MINUS THE BEST COMPETITOR (PERCENTAGE POINTS), WITH POSITIVE/NEGATIVE VALUES IN BOLD RED/BLUE.

| Dataset | Method | LR=0.03 | LR=0.06 | LR=0.09 | LR=0.12 | LR=0.15 |
|---|---|---|---|---|---|---|
| CIFAR100 | CCL | 33.30 | 47.33 | 52.11 | 50.05 | 50.11 |
| | CPE | 23.79 | 38.38 | 42.91 | 41.99 | 41.45 |
| | SimPro | 26.84 | 38.04 | 44.87 | 45.99 | 45.11 |
| | Meta-Expert | 21.98 | 39.53 | 52.29 | 49.22 | 50.23 |
| | Co-GCN | 19.03 | 20.51 | 12.47 | 21.25 | 21.49 |
| | DSRL | - | - | - | - | - |
| | GEGCN | - | - | - | - | - |
| | IMvGCN | 32.95 | 39.72 | 40.43 | 42.65 | 43.98 |
| | MMatch | 13.72 | 14.13 | 13.72 | 13.79 | 13.56 |
| | SPamCo | 18.26 | 25.10 | 23.89 | 35.38 | 40.45 |
| | TMLC | 40.88 | 45.93 | 47.11 | 48.63 | 49.06 |
| | Ours | **43.24** | **54.38** | **58.94** | **62.87** | **64.86** |
| | Δ | **+2.36** | **+7.05** | **+6.65** | **+12.82** | **+14.63** |
| GRAZ02 | CCL | 38.04 | 36.56 | 35.75 | 33.74 | 33.06 |
| | CPE | 35.08 | 29.97 | 31.45 | 32.80 | 34.01 |
| | SimPro | 38.71 | 33.20 | 38.84 | 30.51 | 41.94 |
| | Meta-Expert | 32.12 | 35.35 | 37.23 | 37.23 | 35.22 |
| | Co-GCN | 22.98 | 26.48 | 29.03 | 23.39 | 28.63 |
| | DSRL | 25.00 | 25.00 | 29.03 | 25.00 | 25.00 |
| | GEGCN | 24.33 | 31.05 | 26.48 | 28.63 | 35.62 |
| | IMvGCN | 37.23 | 30.78 | 38.44 | 34.68 | 41.13 |
| | MMatch | 26.08 | 25.00 | 24.87 | 24.87 | 26.75 |
| | SPamCo | 25.00 | 25.00 | 25.00 | 25.27 | 25.00 |
| | TMLC | 34.35 | 31.40 | 32.88 | 28.04 | 37.98 |
| | Ours | **40.73** | **40.05** | **42.88** | **41.13** | **50.00** |
| | Δ | **+2.02** | **+3.49** | **+4.04** | **+3.90** | **+8.06** |
| NoisyMNIST | CCL | 70.53 | 77.98 | 80.83 | 81.33 | 81.97 |
| | CPE | 50.32 | 61.90 | 66.97 | 67.84 | 63.65 |
| | SimPro | 79.74 | 82.69 | 86.17 | 90.39 | 84.45 |
| | Meta-Expert | 79.62 | 81.12 | 75.03 | 86.71 | 84.82 |
| | Co-GCN | 38.25 | 38.07 | 25.32 | 38.54 | 38.28 |
| | IMvGCN | 74.43 | 78.94 | 80.43 | 81.27 | 80.87 |
| | MMatch | 82.28 | 88.23 | 89.46 | 90.81 | 90.97 |
| | SPamCo | 38.34 | 50.01 | 70.02 | 70.82 | 83.42 |
| | TMLC | 63.82 | 67.42 | 67.85 | 69.48 | 71.09 |
| | Ours | **96.95** | **97.44** | **97.34** | **97.45** | **97.63** |
| | Δ | **+14.67** | **+9.21** | **+7.88** | **+6.64** | **+6.66** |
| OutScene | CCL | 54.86 | 65.46 | 63.94 | 61.06 | 59.21 |
| | CPE | 42.55 | 46.63 | 35.34 | 47.68 | 46.23 |
| | SimPro | 44.87 | 48.56 | 56.41 | 53.77 | 50.56 |
| | Meta-Expert | 55.61 | 67.63 | 70.19 | 68.19 | 61.38 |
| | Co-GCN | 20.19 | 21.79 | 20.31 | 19.47 | 20.75 |
| | DSRL | 24.76 | 25.96 | 25.00 | 26.92 | 23.80 |
| | GEGCN | 44.79 | 54.17 | 67.31 | 74.12 | 69.95 |
| | IMvGCN | **61.78** | 67.15 | 70.67 | 70.91 | 67.95 |
| | MMatch | 53.85 | 65.71 | 70.11 | 70.67 | 72.36 |
| | SPamCo | 12.50 | 12.50 | 12.50 | 12.50 | 13.94 |
| | TMLC | 49.12 | 56.97 | 62.34 | 65.38 | 60.18 |
| | Ours | 61.70 | **67.71** | **73.72** | **76.84** | **74.36** |
| | Δ | **-0.08** | **+0.08** | **+3.05** | **+2.72** | **+2.00** |

TABLE VI

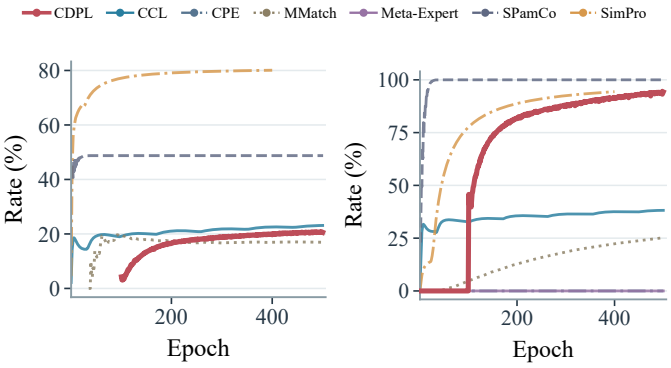
G-MEAN (%) UNDER THE CONSISTENT LONG-TAIL SETTING ($IR = 0.1$) ACROSS LABELING RATIOS $LR \in \{0.03, 0.06, 0.09, 0.12, 0.15\}$. RESULTS ARE MEAN ± STANDARD DEVIATION OVER THREE SEEDS; "—" INDICATES OUT-OF-MEMORY RUNS. BEST AND SECOND-BEST RESULTS ARE MARKED IN PINK/BOLD AND BLUE/UNDERLINED, RESPECTIVELY. Δ DENOTES CDPL MINUS THE BEST COMPETITOR (PERCENTAGE POINTS), WITH POSITIVE/NEGATIVE VALUES IN BOLD RED/BLUE.

| Dataset | Method | LR=0.03 | LR=0.06 | LR=0.09 | LR=0.12 | LR=0.15 |
|---|---|---|---|---|---|---|
| CIFAR100 | CCL | 57.51 | 68.61 | 72.01 | 70.57 | 70.61 |
| | CPE | 48.59 | 61.75 | 65.32 | 64.61 | 64.18 |
| | SimPro | 51.55 | 61.47 | 66.80 | 67.63 | 66.97 |
| | Meta-Expert | 46.56 | 62.61 | 72.11 | 69.95 | 70.68 |
| | Co-GCN | 43.44 | 45.10 | 33.29 | 45.92 | 46.18 |
| | DSRL | - | - | - | - | - |
| | GEGCN | - | - | - | - | - |
| | IMvGCN | 57.21 | 62.83 | 63.39 | 65.12 | 66.13 |
| | MMatch | 24.51 | 22.81 | 20.03 | 20.58 | 19.93 |
| | SPamCo | 42.55 | 49.91 | 47.21 | 59.29 | 63.41 |
| | TMLC | 63.75 | 67.59 | 68.45 | 69.55 | 69.86 |
| | Ours | **65.57** | **73.57** | **76.62** | **79.14** | **80.39** |
| | Δ | **+1.82** | **+4.96** | **+4.51** | **+8.57** | **+9.71** |
| GRAZ02 | CCL | 54.94 | 53.68 | 53.00 | 51.26 | 50.68 |
| | CPE | 52.42 | 47.90 | 49.06 | 50.44 | 51.50 |
| | SimPro | 55.49 | 50.78 | 55.59 | 48.41 | 58.15 |
| | Meta-Expert | 49.82 | 52.58 | 54.23 | 54.25 | 52.45 |
| | Co-GCN | 41.22 | 44.66 | 47.03 | 41.55 | 46.69 |
| | DSRL | 43.30 | 43.30 | 47.08 | 43.30 | 43.30 |
| | GEGCN | 42.63 | 48.89 | 44.68 | 46.71 | 52.85 |
| | IMvGCN | 54.26 | 48.62 | 55.27 | 52.07 | 57.49 |
| | MMatch | 7.76 | 0.00 | 0.00 | 4.52 | 10.84 |
| | SPamCo | 43.30 | 43.30 | 43.30 | 43.56 | 43.30 |
| | TMLC | 50.10 | 47.71 | 48.91 | 44.93 | 52.97 |
| | Ours | **57.13** | **56.61** | **58.91** | **57.48** | **64.55** |
| | Δ | **+1.64** | **+2.93** | **+3.32** | **+3.23** | **+6.40** |
| NoisyMNIST | CCL | 82.59 | 87.22 | 88.94 | 89.24 | 89.62 |
| | CPE | 68.94 | 76.98 | 80.32 | 80.87 | 78.14 |
| | SimPro | 88.28 | 90.05 | 92.09 | 94.55 | 91.07 |
| | Meta-Expert | 88.22 | 89.12 | 85.23 | 92.42 | 91.32 |
| | Co-GCN | 59.69 | 59.54 | 46.54 | 59.93 | 59.71 |
| | IMvGCN | 85.04 | 87.80 | 88.70 | 89.20 | 88.97 |
| | MMatch | 89.18 | 93.14 | 93.92 | 94.70 | 94.79 |
| | SPamCo | 59.76 | 68.72 | 82.27 | 82.78 | 90.49 |
| | TMLC | 78.27 | 80.61 | 80.88 | 81.93 | 82.95 |
| | Ours | **98.29** | **98.57** | **98.51** | **98.58** | **98.68** |
| | Δ | **+9.11** | **+5.43** | **+4.59** | **+3.88** | **+3.89** |
| OutScene | CCL | 70.11 | 78.89 | 77.88 | 75.93 | 74.68 |
| | CPE | 62.37 | 65.62 | 54.80 | 66.37 | 65.32 |
| | SimPro | 64.28 | 67.03 | 72.66 | 70.77 | 68.40 |
| | Meta-Expert | 72.16 | 79.60 | 81.97 | 80.66 | 76.15 |
| | Co-GCN | 42.29 | 44.00 | 42.41 | 41.51 | 42.89 |
| | DSRL | 47.01 | 48.18 | 47.25 | 49.10 | 46.05 |
| | GEGCN | 64.13 | 71.14 | 80.04 | 84.49 | 81.82 |
| | IMvGCN | **76.42** | 80.00 | 82.29 | 82.44 | 80.52 |
| | MMatch | 69.11 | 77.02 | 80.46 | 81.04 | 81.65 |
| | SPamCo | 33.07 | 33.07 | 33.07 | 33.07 | 34.97 |
| | TMLC | 67.47 | 73.12 | 76.80 | 78.84 | 75.33 |
| | Ours | 76.37 | **80.35** | **84.21** | **86.19** | **84.64** |
| | Δ | **-0.05** | **+0.35** | **+1.92** | **+1.70** | **+2.82** |

TABLE VII

SPECIFICITY (%) UNDER THE CONSISTENT LONG-TAIL SETTING ($IR = 0.1$) ACROSS LABELING RATIOS $LR \in \{0.03, 0.06, 0.09, 0.12, 0.15\}$. RESULTS ARE MEAN ± STANDARD DEVIATION OVER THREE SEEDS; "—" INDICATES OUT-OF-MEMORY RUNS. BEST AND SECOND-BEST RESULTS ARE MARKED IN PINK/BOLD AND BLUE/UNDERLINED, RESPECTIVELY. Δ DENOTES CDPL MINUS THE BEST COMPETITOR (PERCENTAGE POINTS), WITH POSITIVE/NEGATIVE VALUES IN BOLD RED/BLUE.

| Dataset | Method | LR=0.03 | LR=0.06 | LR=0.09 | LR=0.12 | LR=0.15 |
|---|---|---|---|---|---|---|
| CIFAR100 | CCL | 99.33 | 99.47 | 99.52 | 99.50 | 99.50 |
| | CPE | 99.23 | 99.38 | 99.42 | 99.41 | 99.41 |
| | SimPro | 99.26 | 99.37 | 99.44 | 99.45 | 99.45 |
| | Meta-Expert | 99.21 | 99.39 | 99.52 | 99.49 | 99.50 |
| | Co-GCN | 99.18 | 99.20 | 99.12 | 99.20 | 99.21 |
| | DSRL | - | - | - | - | - |
| | GEGCN | - | - | - | - | - |
| | IMvGCN | 99.32 | 99.39 | 99.40 | 99.42 | 99.43 |
| | MMatch | 99.13 | 99.13 | 99.13 | 99.13 | 99.13 |
| | SPamCo | 99.17 | 99.24 | 99.23 | 99.35 | 99.40 |
| | TMLC | 99.40 | 99.45 | 99.47 | 99.48 | 99.49 |
| | Ours | **99.43** | **99.54** | **99.59** | **99.62** | **99.65** |
| | Δ | **+0.03** | **+0.07** | **+0.07** | **+0.12** | **+0.15** |
| GRAZ02 | CCL | 79.35 | 78.85 | 78.58 | 77.91 | 77.69 |
| | CPE | 78.36 | 76.66 | 77.15 | 77.60 | 78.00 |
| | SimPro | 79.57 | 77.73 | 79.61 | 76.84 | 80.65 |
| | Meta-Expert | 77.37 | 78.45 | 79.08 | 79.08 | 78.41 |
| | Co-GCN | 74.33 | 75.49 | 76.34 | 74.46 | 76.21 |
| | DSRL | 75.00 | 75.00 | 76.34 | 75.00 | 75.00 |
| | GEGCN | 74.78 | 77.02 | 75.49 | 76.21 | 78.54 |
| | IMvGCN | 79.08 | 76.93 | 79.48 | 78.23 | 80.38 |
| | MMatch | 75.36 | 75.00 | 74.96 | 74.96 | 75.58 |
| | SPamCo | 75.00 | 75.00 | 75.00 | 75.09 | 75.00 |
| | TMLC | 78.45 | 77.47 | 77.96 | 76.35 | 79.66 |
| | Ours | **80.24** | **80.02** | **80.96** | **80.38** | **83.33** |
| | Δ | **+0.67** | **+1.17** | **+1.35** | **+1.30** | **+2.68** |
| NoisyMNIST | CCL | 96.73 | 97.55 | 97.87 | 97.93 | 98.00 |
| | CPE | 94.48 | 95.77 | 96.33 | 96.43 | 95.96 |
| | SimPro | 97.75 | 98.08 | 98.46 | 98.93 | 98.27 |
| | Meta-Expert | 97.74 | 97.90 | 97.23 | 98.52 | 98.31 |
| | Co-GCN | 93.14 | 93.12 | 91.70 | 93.17 | 93.14 |
| | IMvGCN | 97.16 | 97.66 | 97.83 | 97.92 | 97.87 |
| | MMatch | 98.03 | 98.69 | 98.83 | 98.98 | 99.00 |
| | SPamCo | 93.15 | 94.45 | 96.67 | 96.76 | 98.16 |
| | TMLC | 95.98 | 96.38 | 96.43 | 96.61 | 96.79 |
| | Ours | **99.66** | **99.72** | **99.70** | **99.72** | **99.74** |
| | Δ | **+1.63** | **+1.03** | **+0.87** | **+0.74** | **+0.74** |
| OutScene | CCL | 93.47 | 95.07 | 94.85 | 94.44 | 94.17 |
| | CPE | 91.79 | 92.38 | 90.76 | 92.53 | 92.32 |
| | SimPro | 92.12 | 92.65 | 93.77 | 93.40 | 92.94 |
| | Meta-Expert | 93.66 | 95.36 | 95.74 | 95.46 | 94.48 |
| | Co-GCN | 88.60 | 88.83 | 88.62 | 88.50 | 88.68 |
| | DSRL | 89.25 | 89.42 | 89.29 | 89.56 | 89.11 |
| | GEGCN | 92.11 | 93.45 | 95.33 | 96.30 | 95.71 |
| | IMvGCN | **94.54** | 95.31 | 95.81 | 95.84 | 95.42 |
| | MMatch | 93.41 | 95.10 | 95.73 | 95.81 | 96.05 |
| | SPamCo | 87.50 | 87.50 | 87.50 | 87.50 | 87.71 |
| | TMLC | 92.73 | 93.85 | 94.62 | 95.05 | 94.31 |
| | Ours | 94.53 | **95.39** | **96.25** | **96.69** | **96.34** |
| | Δ | **-0.01** | **+0.03** | **+0.44** | **+0.39** | **+0.29** |




(a) Error rate (b) Utilization rate

Fig. 3. Pseudo-label error rate and utilization rate during training on CIFAR-10 at $LR = 0.09$ and $IR = 0.1$. CDPL maintains a low error rate while retaining a high utilization rate.

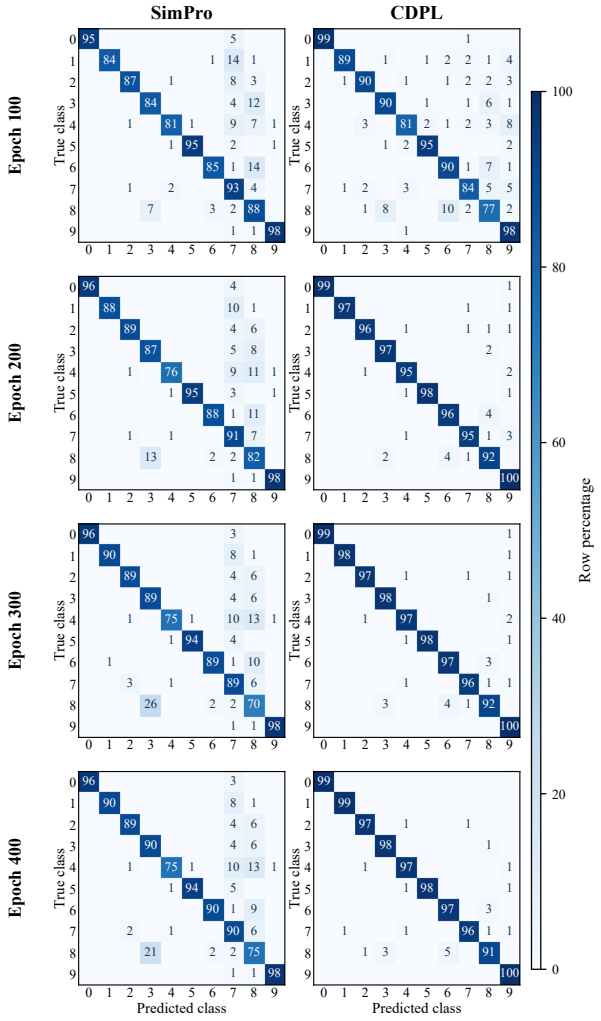
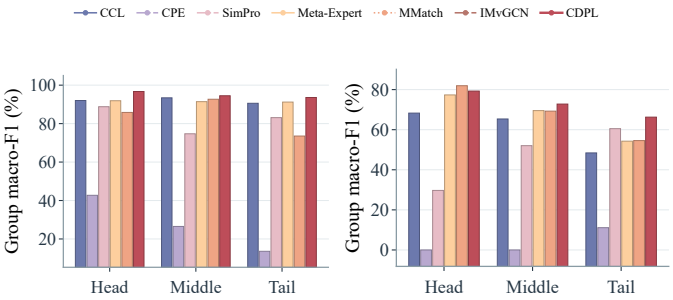
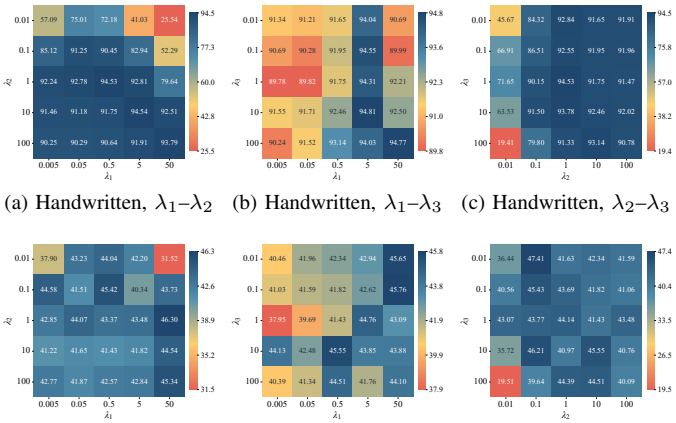



Fig. 4. Row-normalized pseudo-label confusion matrices of SimPro and CDPL on NoisyMNIST at $LR = 0.09$ and $IR = 0.1$.




(a) Handwritten (b) OutScene

Fig. 5. Grouped class-wise Macro-F1 across head, medium, and tail classes on Handwritten and OutScene under the consistent long-tailed setting with $LR = 0.09$ and $IR = 0.1$.



(a) Handwritten, $\lambda_1$–$\lambda_2$ (b) Handwritten, $\lambda_1$–$\lambda_3$ (c) Handwritten, $\lambda_2$–$\lambda_3$

(d) GRAZ02, $\lambda_1$–$\lambda_2$ (e) GRAZ02, $\lambda_1$–$\lambda_3$ (f) GRAZ02, $\lambda_2$–$\lambda_3$

Fig. 6. Parameter analysis of the trade-off coefficients $\lambda_1$, $\lambda_2$, and $\lambda_3$ on Handwritten and GRAZ02.

under four unlabeled distributions (Consistent, Inverse, Arbitrary, and Uniform), covering both training and test sets ($LR = 0.09$, $IR = 0.1$). Across all settings, CDPL produces compact, well-separated class clusters, consistent with its strong and stable classification performance.

## VIII. Additional Clinical Results on the IGF-1 Dataset

The main paper reports four metrics (Accuracy, Macro-F1, G-mean, and Specificity) for the pediatric IGF-1 clinical obesity-status classification task. Figure 9 reports the remaining five metrics, Precision, Recall, macro-averaged AUC (one-vs-rest), macro-averaged AUC (one-vs-one), and macro-averaged Average Precision (AP), under the same labeling ratios $LR \in \{0.06, 0.09, 0.12\}$. CDPL attains the best mean value on all five additional metrics, confirming that its advantage on the clinical task is not limited to the four metrics shown in the main paper.



(a) Consistent–Train (b) Inverse–Train (c) Arbitrary–Train (d) Uniform–Train



(e) Consistent–Test (f) Inverse–Test (g) Arbitrary–Test (h) Uniform–Test

Fig. 7. t-SNE visualization of the fused representations of CDPL on NoisyMNIST under four unlabeled data distributions (consistent, inverse, arbitrary, and uniform), with $LR = 0.09$ and $IR = 0.1$.

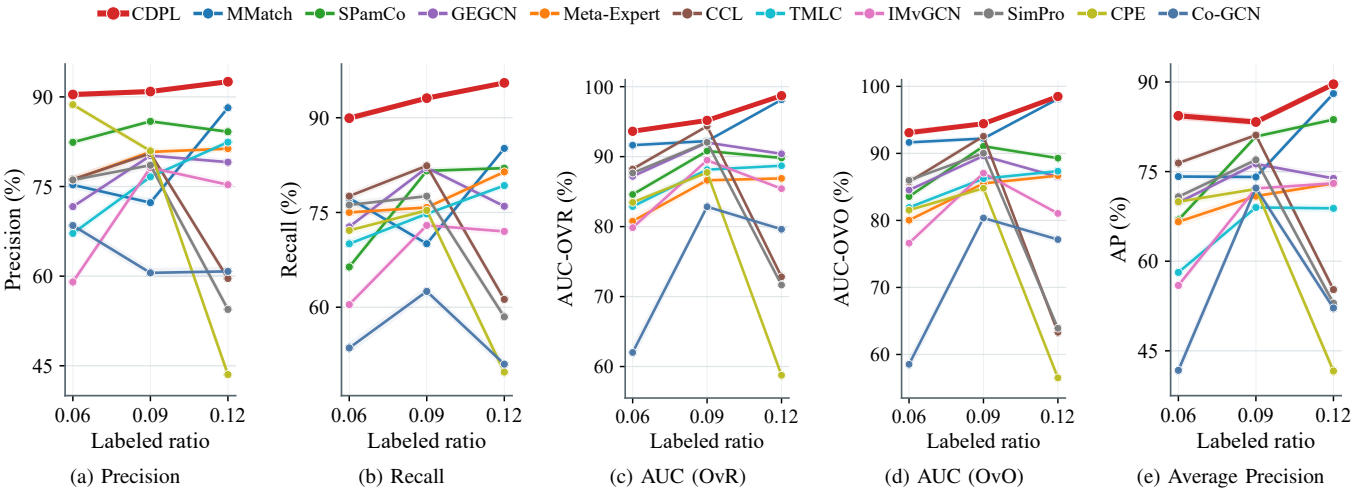


(a) Consistent–Train (b) Inverse–Train (c) Arbitrary–Train (d) Uniform–Train



(e) Consistent–Test (f) Inverse–Test (g) Arbitrary–Test (h) Uniform–Test

Fig. 8. t-SNE visualization of the fused representations of CDPL on YouTubeFace20 under four unlabeled data distributions (consistent, inverse, arbitrary, and uniform), with $LR = 0.09$ and $IR = 0.1$.




(a) Precision (b) Recall (c) AUC (OvR) (d) AUC (OvO) (e) Average Precision

Fig. 9. Precision, Recall, macro-averaged AUC (one-vs-rest and one-vs-one), and macro-averaged Average Precision on the pediatric IGF-1 dataset at $LR \in \{0.06, 0.09, 0.12\}$.